Step 1: Project Revenues

The principal business activities of most firms involve generating revenues by selling products or delivering services. Analysts frequently use the expected future level of revenues as a basis for deriving many other amounts in the financial statement forecasts.[3] Therefore, analysts typically begin the forecasting process by projecting revenues.

Revenues are determined by sales volumes (quantities) and prices. Some firms report sales volumes (for example, automobile manufacturers report numbers of vehicles sold and beverage makers report gallons or cases sold), enabling you to assess volume and price separately as drivers of past revenues and predictors of future revenues. Some firms report volume-related measures that you can use to forecast revenues, such as the number of stores for retailers and restaurant chains or the number of passenger and revenue seat miles for airlines. When you perform the industry analysis in the first step of the financial statement analysis process, you will discover industrywide conditions that will influence your volume forecasts. For a stable firm in a mature industry, you might conclude that the firm will not significantly gain or lose market share, but that sales volume will grow with the population in the firm's geographic markets. For a firm that has increased its production capacity in an industry with high anticipated growth (for example, biotechnology or tablet computers), you can use the industry growth rate coupled with the expansion in the firm's capacity to project sales volume increases.

When projecting *prices*, which is difficult, you should consider factors specific to the firm and its industry that will affect demand and price elasticity, such as excess or constrained capacity, raw material surpluses or shortages, substitute products, and technological changes in products or production methods. Capital-intensive firms, such as manufacturers of paper products or computer chips, may require several years to add new capacity. If the firm competes in a capital-intensive industry that you expect will operate near capacity for the next few years, price increases will be more likely. By contrast, if the firm competes in a capital-intensive industry with excess capacity, price increases will be less likely. Further, capital-intensive firms in a competitive industry with excess capacity may face high exit barriers and thus may experience future price decreases. A firm with significant technological improvements in its production processes (for example, some portions of the computer industry or cell phone industry) might expect increases in sales volume but decreases in prices. If a firm has established a brand name or unique characteristics for its products, it may have a greater potential to maintain or increase prices than a competitor with generic products.

You also should consider economywide factors such as the expected rate of inflation and changes in foreign currency exchange rates. If the firm generates some of its revenues in a foreign currency that has appreciated relative to the currency used to prepare the firm's financial statements, the translation process will increase reported revenues due to the foreign exchange rate increase, thus leading to a higher revenue growth unrelated to volume or pricing increases. In addition, your revenue forecasts should also capture the effects of corporate transactions such as acquisitions and divestitures. Acquisitions typically increase revenues, whereas divestitures of subsidiaries usually reduce revenues.

[3] Revenue forecasts are particularly important because they have crucial impacts on the projected income statements, balance sheets, and cash flows. You should utilize as much relevant information as possible in developing reliable revenue forecast assumptions. You can often find useful information for forecasts in company disclosures, competitors' financial statements and disclosures, industry data, and regional- and country-specific economic data. This chapter seeks to illustrate the techniques of using such information in developing forecasts, while also being concise and efficient for you, the reader.

If revenues have grown at a reasonably steady rate in prior periods and you expect that economic, industry, and firm-specific factors will remain stable, you can project that the historical growth rate will persist into the future. Projecting revenues for a cyclical firm (for example, heavy machinery manufacturers, property-casualty insurers, and investment banks) involves an additional degree of difficulty because revenue growth rates often exhibit wide variations in both direction and amount over the business cycle. For such firms, you can project revenue growth rates that vary with the stages of the cycle, assuming you can correctly identify the firm's current point in the cycle and how quickly it will progress to the next stages.

It is challenging to project future revenues for firms that depend heavily on new products to generate growth (such as pharmaceutical firms conducting research to discover new drugs, technology firms developing new products, or oil and gas firms exploring for new reserves) because of the inherent uncertainty in the discovery and development process. In those cases, analysts often rely heavily on firms' disclosures about the research and development pipeline. In addition, analysts often seek help from scientists, doctors, or technology experts who can provide knowledgeable advice on the likelihood of the future success of the firm's research and product development efforts.

This discussion reinforces how forecasting depends heavily on the first four steps of the analysis process. Projecting a firm's future business activities, particularly revenues, relies on understanding the economic and competitive forces of the industry, the competitive strategy of the firm, the quality of the firm's accounting, and the drivers of the firm's profitability and risk.

Projecting Revenues for Starbucks

Earlier chapters indicated that the coffee shop industry is very mature and competitive. **Starbucks** has established a very strong niche in this industry as the world's largest chain of coffee shops. It has generated impressive revenue growth, with a compounded annual growth rate of 13.5% from 2013 to 2015.

In **Starbucks'** 2015 Annual Report, the MD&A section titled "Results of Operations" discloses information about revenues and operating profits for each of **Starbucks'** five operating segments. The largest revenue-generating segment is the Americas segment, which consists of company-operated and licensed stores in North America, South America, and Central America. The second-largest segment in terms of revenues is the China/Asia Pacific (CAP) segment, which consists of company-operated and licensed stores in Asia, especially China and Japan. The EMEA segment consists mainly of company-operated and licensed stores in Europe, the Middle East, and Africa. The fourth segment is Channel Development, which consists of sales of coffee beans and other products to grocery stores and foodservice distributors. **Starbucks** also groups other small business it controls into the fifth and smallest segment, the All Other segment. It primarily includes company-operated stores under the Teavana brand.

For each segment, **Starbucks** discloses revenues, operating expenses, and operating income. **Starbucks** also discloses overall revenue growth rates, revenue growth attributable to opening new stores, comparable store sales growth rates (arising from changes in transaction volume and prices), as well as effects of changes in foreign exchange rates and acquisitions and divestitures. The data for **Starbucks'** revenue drivers appear in Exhibit 10.2. These data reveal significant differences in the drivers of growth across the five segments. For example, the Americas segment generated 11% sales growth in 2015 primarily from company-operated stores and comparable store sales growth rates. The CAP segment experienced revenue growth of 112% in 2015, in large part due to

Exhibit 10.2

Starbucks' Revenue Growth Analysis by Segment (dollar amounts in millions)

Year	2013	2014	2015
Net Revenues (in millions)			
Company-operated stores	$11,793.2	$12,977.9	$15,197.3
Licensed stores	1,360.5	1,588.6	1,861.9
CPG, foodservice and other	1,713.1	1,881.3	2,103.5
Total Net Revenues	$14,866.8	$16,447.8	$19,162.7
Growth rates		10.6%	16.5%
Americas	**2013**	**2014**	**2015**
Net new stores opened during the year			
Company-operated	276	317	276
Licensed	404	381	336
Total	680	698	612
Total stores			
Company-operated	8,078	8,395	8,671
Licensed	5,415	5,796	6,132
Total	13,493	14,191	14,803
Revenues			
Company-operated	$ 10,038.3	$ 10,866.5	$ 11,925.6
Growth rates		8.3%	9.7%
Revenues per store/year	$ 1.264	$ 1.319	$ 1.398
Growth rates		4.4%	5.9%
Licensed	$ 915.4	$ 1,074.9	$ 1,334.4
Growth rates		17.4%	24.1%
Revenues per store/year	$ 0.176	$ 0.192	$ 0.224
Growth rates		9.2%	16.7%
CAP	**2013**	**2014**	**2015**
Net new stores opened during the year			
Company-operated	239	250	1,320
Licensed	349	492	(482)
Total	588	742	838
Total stores			
Company-operated	882	1,132	2,452
Licensed	3,000	3,492	3,010
Total	3,882	4,624	5,462

(Continued)

Exhibit 10.2 (Continued)			
CAP	**2013**	**2014**	**2015**
Revenues			
Company-operated	$ 671.7	$ 859.4	$ 2,127.3
Growth rates		27.9%	147.5%
Revenues per store/year	$ 0.881	$ 0.853	$ 0.926
Growth rates		(3.1%)	8.5%
Licensed	$ 245.3	$ 270.2	$ 264.4
Growth rates		10.2%	(2.1%)
Revenues per store/year	$ 0.087	$ 0.083	$ 0.096
Growth rates		(4.1%)	15.6%
EMEA	**2013**	**2014**	**2015**
Net new stores opened during the year			
Company-operated	(29)	(9)	(80)
Licensed	129	180	302
Total	100	171	222
Total stores			
Company-operated	826	817	737
Licensed	1,143	1,323	1,625
Total	1,969	2,140	2,362
Revenues			
Company-operated	$ 932.8	$ 1,013.8	$ 911.2
Growth rates		8.7%	(10.1%)
Revenues per store/year	$ 1.110	$ 1.234	$ 1.173
Growth rates		11.2%	(5.0%)
Licensed	$ 190.3	$ 238.4	$ 257.2
Growth rates		25.3%	7.9%
Revenues per store/year	$ 0.176	$ 0.193	$ 0.174
Growth rates		9.6%	(9.8%)
Other Stores (including Teavana)	**2013**	**2014**	**2015**
Net new stores opened during the year			
Company-operated	343	12	6
Licensed	(10)	(24)	(1)
Total	333	(12)	5

Exhibit 10.2 (Continued)			
Other Stores (including Teavana)	**2013**	**2014**	**2015**
Other segment stores			
Company-operated	357	369	375
Licensed	66	42	41
Total	423	411	416
Other segment store revenues	$ 159.9	$ 243.3	$ 239.1
Growth rates		52.2%	(1.7%)
CPG, Foodservice, and Other	**2013**	**2014**	**2015**
Revenues	$ 1,713.1	$ 1,881.3	$ 2,103.5
Growth rates		9.8%	11.8%
Total Net Revenues	$14,866.8	$16,447.8	$19,162.7

Starbucks gaining ownership and control of 1,009 company-operated stores by acquiring **Starbucks** Japan. By contrast, the EMEA segment experienced a 6% decline in revenues during 2015, in part due to unfavorable foreign currency exchange rates and net closures of some company-operated stores.

In addition, in **Starbucks'** 2015 Annual Report, the MD&A section titled "Fiscal 2016—The View Ahead" discloses information from **Starbucks'** managers about their expectations. **Starbucks'** managers update those expectations with each quarterly earnings release and quarterly report during fiscal 2016. In these disclosures, **Starbucks** reveals it expects to open roughly 1,800 new stores during fiscal 2016. Of these new stores, **Starbucks** expects to open 700 new stores in the Americas, with roughly half of them being company operated and the other half being licensed stores. **Starbucks** also expects to open 900 new stores in the CAP segment, with roughly one-third being company operated and the other two-thirds being licensed stores. In the EMEA segment, **Starbucks** expects to open 200 net new stores, primarily licensed stores. **Starbucks** does not disclose any plans to open any stores in the All Other segment. **Starbucks** also discloses that it expects comparable store sales growth rates to be "somewhat above mid-single digits."

We analyze the drivers of revenue growth at the segment level by store type (company operated versus licensed) in order to develop more accurate forecasts for **Starbucks'** revenue growth in each segment. In addition, we will consider **Starbucks'** disclosures about expected future store openings and growth rates in 2016. These disclosures are worth considering because in the past **Starbucks** has provided reasonably accurate disclosures about its expected future activities. After we develop our forecast assumptions, we will then discuss and apply an adjustment to our forecast amount for fiscal 2016, which will contain 53 weeks rather than the normal 52 weeks.

Americas Revenue Growth

The Americas segment generates revenues from operating company stores and from licensed stores in the Western Hemisphere—North, South, and Central America. The data in Exhibit 10.2 reveal that company-operated store revenues in the Americas grew by 8.3% in 2014 and by 9.7% in 2015. **Starbucks** discloses that it achieved these growth rates through a combination of opening new stores and same-store sales growth. **Starbucks** opened 317 new company stores in the Americas in 2014 and 276 new stores in 2015. The average sales per store grew by 4.4% to $1.319 million in 2014, followed by an additional 5.9% growth, to $1.398 million, in 2015.

The data in Exhibit 10.2 also reveal that licensed-store revenues in the Americas grew by 17.4% in 2014 and by 24.1% in 2015. **Starbucks** discloses that it achieved these impressive growth rates through a combination of opening new stores and same-store sales growth. **Starbucks** opened 381 new licensed stores in the Americas in 2014 and 336 new licensed stores in 2015. The average revenues per licensed store grew by 9.2% to $0.192 million in 2014, followed by an additional 16.7% growth, to $0.224 million, in 2015. [Recall that **Starbucks'** licensed store revenues are much lower per licensed store compared to company-operated stores because they only consist of royalties and license fees, as well as product sales to licensees.]

In **Starbucks'** 2015 Annual Report, the MD&A section titled "Fiscal 2016—The View Ahead" and in subsequent quarterly reports during fiscal 2016, **Starbucks** discloses that it expects to open roughly 350 net new company-operated stores and roughly 350 net new licensed stores. In prior years, similar projections by **Starbucks** have turned out to be reasonably accurate. These projections are in line with **Starbucks'** recent store openings in the Americas, and the firm seems to have the financial capital and management competence to handle this level of growth. In our forecast assumptions, we will use these projections for new store openings in Year +1 (fiscal 2016). We will also assume that **Starbucks** will sustain the same level of new store openings in Year +2 through Year +5.

As noted earlier, in the 2015 Annual Report, **Starbucks** also discloses that it expects comparable store sales growth rates to be "somewhat above mid-single digits." For stores in the Americas segment, we project that average revenues for company-operated and licensed stores will therefore grow by 6% per year.

We compute future revenue amounts as the average number of stores open during the year times the average revenue per store. So for company-operated stores, **Starbucks** began Year +1 with 8,671 stores in the Americas. If it opens 350 stores during that year, the result will be 9,021 company-operated stores by year-end. If the new stores are open for roughly one-half of the year on average, then there will be roughly 8,846 store-years [(8,671 + 9,021)/2] generating revenues during Year +1. Average revenues per store in 2015 were $1.398 million. If store revenues grow by 6%, then the average store will generate revenues of $1.481 million [$1.398 million × 1.06] during Year +1. Therefore, we project that during Year +1 **Starbucks'** company-operated stores in the Americas segment will generate total revenues amounting to $13,104.8 million [8,846 store-years × $1.481 million average revenues per store-year]. We use similar computations to project revenue amounts for Years +2 to +5.

For licensed stores, **Starbucks** began Year +1 with 6,132 stores in the Americas. If it opens 350 stores during that year, the result will be 6,482 licensed stores by year-end. If the new stores are open for roughly one-half of the year on average, then there will be roughly 6,307 store-years [(6,132 + 6,482)/2] generating revenues during Year +1. Average revenues per store in 2015 were $0.224 million. If licensed store revenues grow by 6%, then the average store will generate revenues of $0.237 million [$0.224 million × 1.06] during Year +1. Therefore, total revenues generated during Year +1 among licensed stores in the Americas segment will be $1,495.8 million [6,307 store-years × $0.237 million average revenues per store-year]. We use similar computations to project revenue amounts for Years +2 to +5.

The revenue projections and growth rates for **Starbucks'** Americas segment over the five-year forecast horizon are as follows (allow for rounding):

Americas	Actuals 2015	Projections Year +1	Year +2	Year +3	Year +4	Year +5
Net new stores during the year						
Company-operated	276	350	350	350	350	350
Licensed	336	350	350	350	350	350
Total	612	700	700	700	700	700
Total stores						
Company-operated	8,671	9,021	9,371	9,721	10,071	10,421
Licensed	6,132	6,482	6,832	7,182	7,532	7,882
Total	14,803	15,503	16,203	16,903	17,603	18,303
Revenues						
Company-operated	$11,925.6	$13,104.8	$14,440.7	$15,889.8	$17,460.7	$19,162.9
Revenues per store/year	$ 1.398	$ 1.481	$ 1.570	$ 1.665	$ 1.764	$ 1.870
Projected growth rates		6.0%	6.0%	6.0%	6.0%	6.0%
Licensed	$ 1,334.4	$ 1,495.8	$ 1,673.5	$ 1,867.2	$ 2,078.1	$ 2,307.6
Revenues per store/year	$ 0.224	$ 0.237	$ 0.251	$ 0.266	$ 0.282	$ 0.299
Projected growth rates		6.0%	6.0%	6.0%	6.0%	6.0%

China/Asia Pacific (CAP) Segment Revenue Growth

The CAP segment generates revenues from company-operated stores and from licensed stores in China, Japan, Korea, and other Asia-Pacific countries. The data in Exhibit 10.2 reveal that company-operated store revenues in the CAP segment grew by 27.9% in 2014 and by 147.5% in 2015. **Starbucks** discloses that it achieved these growth rates through a combination of opening new stores, same-store sales growth, and acquisition. In particular, **Starbucks** acquired 1,009 company-operated stores in Japan by acquiring 100% ownership of a former joint venture partner, **Starbucks** Japan. These stores in Japan had been reported as licensed stores before the acquisition and are company-operated stores after the acquisition. In addition, **Starbucks** also discloses that it opened 250 new company stores in the CAP segment in 2014 and 311 new stores in 2015. The average sales per store fell by 3.1% to $0.853 million in 2014, followed by 8.5% growth, to $0.926 million in 2015.[4]

The data in Exhibit 10.2 also reveal that licensed store revenues in the CAP segment grew by 10.2% in 2014 and then fell by 2.1% in 2015. **Starbucks** opened 492 new licensed stores in 2014 and 527 new licensed stores in 2015 (not including the 1,009 stores in Japan that transitioned from licensed to company operated). The average revenues per licensed store fell by 4.1% to $0.083 million in 2014, and then grew by 15.6%, to $0.096 million in 2015.

In **Starbucks'** 2015 Annual Report, the MD&A section titled "Fiscal 2016—The View Ahead," and in subsequent quarterly reports during fiscal 2016, **Starbucks** discloses that it expects to open roughly 300 net new company-operated stores and roughly 600 net new licensed stores in the CAP segment. These projections are in line with **Starbucks'** recent store openings in the CAP region, and the firm seems to have the financial capital and management competence to handle this level of growth. In our forecast assumptions, we will use these projections for new store openings in Year +1 (fiscal 2016). We will also assume that **Starbucks** will sustain the same level of new store openings in Year +2 through Year +5.

As noted earlier, in the 2015 Annual Report, **Starbucks** also discloses that it expects comparable store sales growth rates to be "somewhat above mid-single digits." For stores in the CAP segment, which is **Starbucks'** fastest-growing segment, we project that average revenues for company-operated and licensed stores will grow by 7% per year.

We again compute future revenue amounts as the average number of stores open during the year times the average revenue per store. So for company-operated stores in the CAP segment, **Starbucks** began Year +1 with 2,452 stores. If it opens 300 stores during that year, the result will be 2,752 company-operated stores by year-end. If the new stores are open for roughly one-half of the year on average, then there will be roughly 2,602 store-years [(2,452 + 2,752)/2] generating revenues during Year +1. Average revenues per store in 2015 were \$0.926 million. If store revenues grow by 7%, then the average store will generate revenues of \$0.991 million [\$0.926 million × 1.07] during Year +1. Therefore, total revenues generated during Year +1 among company-operated stores in the CAP segment will be \$2,579.0 million [2,602 store-years × \$0.991 million average revenues per store-year]. We use similar computations to project revenue amounts for Years +2 to +5.

For licensed stores, **Starbucks** began Year +1 with 3,010 stores in the CAP segment. If it opens 600 stores during that year, the result will be 3,610 licensed stores by year-end. If the new stores are open for roughly one-half of the year on average, then there will be roughly 3,310 store-years [(3,010 + 3,610)/2] generating revenues during Year +1. Average revenues per store in 2015 were \$0.096 million. If licensed store revenues grow by 7%, then the average store will generate revenues of \$0.103 million [\$0.096 million × 1.07] during Year +1. Therefore, total revenues generated during Year +1 among licensed stores in the CAP segment will be \$341.0 million [3,310 store-years × \$0.103 million average revenues per store-year]. We use similar computations to project revenue amounts for Years +2 to +5.

The revenue projections and growth rates for **Starbucks'** CAP segment over the five-year forecast horizon are as follows (allow for rounding):

	Actuals	Projections				
CAP	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Net new stores during the year						
Company-operated	276	300	300	300	300	300
Licensed	336	600	600	600	600	600
Total	612	900	900	900	900	900
Total stores						
Company-operated	2,452	2,752	3,052	3,352	3,652	3,952
Licensed	3,010	3,610	4,210	4,810	5,410	6,010
Total	5,462	6,362	7,262	8,162	9,062	9,962
Revenues						
Company-operated	\$2,127.3	\$2,579.0	\$3,077.7	\$3,633.6	\$4,252.2	\$4,939.6
Revenues per store/year	\$ 0.926	\$ 0.991	\$ 1.061	\$ 1.135	\$ 1.214	\$ 1.299
Projected growth rates		7.0%	7.0%	7.0%	7.0%	7.0%
Licensed	\$ 264.4	\$ 341.0	\$ 430.9	\$ 531.9	\$ 644.8	\$ 771.0
Revenues per store/year	\$ 0.096	\$ 0.103	\$ 0.110	\$ 0.118	\$ 0.126	\$ 0.135
Projected growth rates		7.0%	7.0%	7.0%	7.0%	7.0%

Europe, Middle East, and Africa (EMEA) Revenue Growth

The EMEA segment generates revenues from company-operated stores and from licensed stores in countries in Europe, the Middle East, and Africa, including the United Kingdom, Turkey, the UAE, Russia, and many others. The data in Exhibit 10.2 reveal that company-operated store revenues in the EMEA segment grew by 8.7% in 2014 but fell by 10.1% in 2015. The average sales per company-operated store in EMEA grew by 11.2% to $1.234 million in 2014, but then fell by 5.0% to $1.173 million in 2015. During 2014, **Starbucks** closed a net 9 stores, and then it closed another net 80 stores in 2015.

The data in Exhibit 10.2 also reveal that licensed-store revenues in the EMEA segment grew by 25.3% in 2014 and then grew by an additional 7.9% in 2015. **Starbucks** opened 180 new licensed stores in 2014 and 302 new licensed stores in 2015. The average revenues per licensed store grew by 9.6% to $0.193 million in 2014, but then fell by 9.8% to $0.174 million in 2015.

In the 2015 Annual Report, **Starbucks** discloses that it expects to open roughly 200 net new licensed stores in the EMEA segment. In our forecast assumptions, we will project no new company-operated store openings but 200 new licensed stores in EMEA during Year +1 (fiscal 2016). We will also assume that **Starbucks** will sustain the same level of new store openings in Year +2 through Year +5.

Comparable store revenue growth rates in the EMEA segment have been lagging **Starbucks'** growth in other segments for several years, so it seems unlikely that the mid-single-digits average growth rate (noted earlier) will apply to EMEA stores. For stores in the EMEA segment, which is **Starbucks'** slowest-growing segment, we project that average revenues will grow by only 2% per year.

We again compute future revenue amounts as the average number of stores open during the year times the average revenue per store. So for company-operated stores in the EMEA segment, with no new store openings, **Starbucks** will begin and end Year +1 with 737 stores. Average revenues per store in 2015 were $1.173 million. If store revenues grow by 2%, then the average store will generate revenues of $1.196 million [$1.173 million × 1.02] during Year +1. Therefore, total revenues generated by company-operated stores in the EMEA segment during Year +1 will be $881.6 million [737 store-years × $1.196 million average revenues per store-year]. We use similar computations to project revenue amounts for Years +2 to +5.[5]

For licensed stores, **Starbucks** began Year +1 with 1,625 stores in the EMEA segment. If it opens 200 stores during that year, the result will be 1,825 licensed stores by year-end. If the new stores are open for roughly one-half of the year on average, then there will be roughly 1,725 store-years [(1,625 + 1,825)/2] generating revenues during Year +1. Average revenues per store in 2015 were $0.174 million. If licensed store revenues grow by 2%, then the average store will generate revenues of $0.178 million [$0.174 million × 1.02] during Year +1. Therefore, total revenues generated by licensed stores in the EMEA segment during Year +1 will be $307.0 million [1,725 store-years × $0.178 million average revenues per store-year]. We use similar computations to project revenue amounts for Years +2 to +5.

The revenue projections and growth rates for **Starbucks'** EMEA segment over the five-year forecast horizon are as follows (allow for rounding):

	Actuals	Projections				
EMEA	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Net new stores during the year						
Company-operated	(80)	0	0	0	0	0
Licensed	302	200	200	200	200	200
Total	222	200	200	200	200	200
Total stores						
Company-operated	737	737	737	737	737	737
Licensed	1,625	1,825	2,025	2,225	2,425	2,625
Total	2,362	2,562	2,762	2,962	3,162	3,362
Revenues						
Company-operated	\$911.2	\$881.6	\$899.2	\$917.2	\$935.5	\$954.2
Revenues per store/year	\$1.173	\$1.196	\$1.220	\$1.244	\$1.269	\$1.295
Projected growth rates		2.0%	2.0%	2.0%	2.0%	2.0%
Licensed	\$257.2	\$307.0	\$349.5	\$393.5	\$439.1	\$486.4
Revenues per store/year	\$0.174	\$0.178	\$0.182	\$0.185	\$0.189	\$0.193
Projected growth rates		2.0%	2.0%	2.0%	2.0%	2.0%

Other Stores Segment Revenue Growth

The Other Stores segment primarily includes company-operated stores operating under the Teavana and Seattle's Best Coffee brands, as well as other businesses. The data in Exhibit 10.2 reveal that revenues in the Other Stores segment grew by 52.2% in 2014 but fell by 1.7% in 2015. Total revenues in the Other Stores segment amounted to \$239.1 million in fiscal 2015. During 2014, **Starbucks** opened only 12 stores, and then it opened another 6 stores in 2015, to reach a total of 375 company-operated stores. The Other Stores segment also includes 41 licensed stores as of the end of 2015, and it has been closing licensed stores over the past couple of years.

In **Starbucks'** 2015 Annual Report, it does not disclose plans to open new stores in the Other Stores segment. In our forecast assumptions, we will simply project no new store openings in the Other Stores segment during Year +1 through Year +5. We will also simply project revenues in this segment will grow by only 3% per year.

The revenue projections and growth rates for **Starbucks'** Other Stores segment over the five-year forecast horizon are as follows (allow for rounding):

	Actuals	Projections				
Other Stores (including Teavana)	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Total stores						
Company-operated	375	375	375	375	375	375
Licensed	41	41	41	41	41	41
Total	416	416	416	416	416	416
Revenues	\$239.1	\$246.3	\$253.7	\$261.3	\$269.1	\$277.2
Projected growth rates		3.0%	3.0%	3.0%	3.0%	3.0%

CPG, Foodservice, and Other Segment Revenue Growth

Starbucks' CPG, Foodservice, and Other segment produces and sells various products through different channels. The CPG (Consumer Packaged Goods) business generates domestic and international sales of packaged coffee and tea, ready-to-drink beverages, and single-serve coffee and tea products through grocery stores, warehouse clubs, and specialty retailers. If you purchase a bag of **Starbucks'** coffee beans from your local grocery store, that store may have purchased them from **Starbucks'** CPG business. The Foodservice business sells coffee beans, tea, and related products to institutional foodservice companies. If you had a cup of **Starbucks'** coffee in a cafeteria, in a restaurant, or on a flight, it is likely that coffee was originally produced and sold by **Starbucks'** Foodservice business.

The data in Exhibit 10.2 reveal that revenues in the CPG, Foodservice, and Other segment have been growing steadily, by 9.8% in 2014 and by 11.8% in 2015. Total revenues in the CPG, Foodservice, and Other segment amounted to $2,103.5 million in fiscal 2015.[6]

In the 2015 Annual Report, **Starbucks** does not disclose specific growth expectations for this segment. In our forecast assumptions, we will simply project that it will sustain roughly 10% growth in revenues per year during Year +1 through Year +5, consistent with recent performance. The revenue projections and growth rates for **Starbucks'** CPG, Foodservice, and Other segment over the five-year forecast horizon are as follows (allow for rounding):

	Actuals	Projections				
CPG, Foodservice, and Other	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Revenues	$2,103.5	$2,313.9	$2,545.2	$2,799.8	$3,079.7	$3,387.7
Projected growth rates		10.0%	10.0%	10.0%	10.0%	10.0%

Starbucks' Combined Revenue Growth Forecasts, Including the 53rd-Week Effect

The following table combines the revenue forecasts for each of **Starbucks'** five segments into the three revenue line-items **Starbucks** reports on its income statements: Company-Operated Stores; Licensed Stores; and CPG, Foodservice, and Other.

Starbucks uses an accounting convention in which each quarter normally consists of 13 weeks, and a fiscal year consists of 52 weeks. However, **Starbucks** also uses an accounting convention in having the last day of the fiscal year fall on the Sunday closest to September 30th. Together, that means that once every six or seven years, **Starbucks** has to add a 53rd week to its fiscal year, so the year-end falls close to September 30th. Our forecast Year +1 (fiscal 2016) will be the next time when **Starbucks'** fiscal year will be 53 weeks instead of only 52 weeks.

It is straightforward to incorporate the 53rd-week effect into our forecasts for Year +1. We simply need to multiply our revenue forecast amounts for that year by a factor of 53/52 (= 1.019). The following table presents the projected revenue amounts for each segment, **Starbucks'** total net revenues, and annual growth rates after including the 53rd-week effect, for each year through Year +5.

Starbucks' Combined Revenue Forecasts

	Actuals	Projections				
Company-Operated Stores	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Americas	$11,925.6	$13,104.8	$14,440.7	$15,889.8	$17,460.7	$19,162.9
CAP	$ 2,127.3	$ 2,579.0	$ 3,077.7	$ 3,633.6	$ 4,252.2	$ 4,939.6
EMEA	$ 911.2	$ 881.6	$ 899.2	$ 917.2	$ 935.5	$ 954.2
Other Stores	$ 239.1	$ 246.3	$ 253.7	$ 261.3	$ 269.1	$ 277.2
Subtotals	$15,203.2	$16,811.7	$18,671.3	$20,701.8	$22,917.5	$25,334.0
The 53rd-week effect (53/52)		1.019				
Totals	$15,203.2	$17,135.0	$18,671.3	$20,701.8	$22,917.5	$25,334.0
Licensed Stores	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Americas	$1,334.4	$1,495.8	$1,673.5	$1,867.2	$2,078.1	$2,307.6
CAP	$ 264.4	$ 341.0	$ 430.9	$ 531.9	$ 644.8	$ 771.0
EMEA	$ 257.2	$ 307.0	$ 349.5	$ 393.5	$ 439.1	$ 486.4
Subtotals	$1,856.0	$2,143.8	$2,454.0	$2,792.6	$3,162.1	$3,565.0
The 53rd-week effect (53/52)		1.019				
Totals	$1,856.0	$2,185.0	$2,454.0	$2,792.6	$3,162.1	$3,565.0
CPG, Foodservice, and Other	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Revenues	$ 2,103.5	$ 2,313.9	$ 2,545.2	$ 2,799.8	$ 3,079.7	$ 3,387.7
The 53rd-week effect (53/52)		1.019				
Totals	$ 2,103.5	$ 2,358.3	$ 2,545.2	$ 2,799.8	$ 3,079.7	$ 3,387.7
Total Net Revenues	$19,162.7	$21,678.3	$23,670.5	$26,294.2	$29,159.4	$32,286.7

The Forecasts spreadsheet in FSAP gives you the opportunity to input specific forecast parameters (such as revenue growth rates) for Year +1 through Year +5, as well as general forecast parameters for Year +6 and beyond. For Years +1 through +5, we will enter the revenue growth rates and amounts shown above. The forecast parameters for Year +6 and beyond represent general forecast assumptions over the long-run horizon. We assume **Starbucks** will sustain a 3.0% nominal growth rate in revenue in Year +6 and beyond, consistent with the sum of the expected long-run growth rate in the economy plus the expected long-run inflation rate averaging 3.0% per year.[7]

You can use the Forecast Development spreadsheet in FSAP to develop detailed revenues forecasts, capturing the key drivers of the firm's growth. We have done that here by analyzing and forecasting **Starbucks'** revenue growth drivers separately for each segment and for each type of store, and then aggregating those forecasts into total revenue forecasts through Year +5. Appendix C illustrates how we used the forecast development spreadsheet to develop the revenue forecasts for **Starbucks**.

Step 2: Project Operating Expenses

The procedure for projecting operating expenses depends on the degree to which they have fixed or variable components. If certain operating expenses vary directly with sales, you can project these future operating expenses by either multiplying projected sales by the appropriate common-size percentage or by projecting those operating expenses to grow at the same rate as sales.

Some operating expenses reflect cost structures that will not change linearly with sales. For example, the firm may experience economies of scale as sales increase or may face expenses that remain relatively fixed even if sales decrease. In cases like those, using the common-size income statement approach will project operating expenses that are too high or too low.

Some operating expenses have both a fixed component and a portion that varies with sales. For example, a firm's selling, general, and administrative expenses often include fixed components for items such as salaries, rent, insurance, and other corporate overhead expenses but variable components that vary directly with sales, such as sales commissions. A possible clue for the existence of fixed costs can be found in the ratio of the percentage change in an expense relative to the percentage change in sales. Changes in this ratio over time may be due to the existence of fixed costs.

When sales grow at faster rates than costs of goods sold or selling, general, and administrative expenses, it often indicates the presence of fixed costs. You can estimate the variable cost component as a percentage of sales by dividing the amount of the change in the expense by the amount of the change in sales for the same period. You can then multiply sales by the variable-cost percentage to estimate the variable cost component. Subtracting the variable cost component from the total expense yields an estimate of the fixed cost for that particular operating expense. For example, suppose sales grow from \$10 million to \$12 million, while costs of sales grow from \$7 million to \$8 million. The \$1 million increase in costs of sales divided by the \$2 million increase in sales implies that variable costs are roughly 50% of sales. If so, it also implies that fixed costs amount to \$2 million per year. In the current year, for example, costs of sales are comprised of \$6 million in variable costs (\$12 million × 50%) plus \$2 million in fixed costs. Using this approach, you can then project a particular future expense with a fixed component and a component that varies with sales.[8]

When projecting operating expenses as a percentage of sales, you should keep in mind that an expense as a percentage of sales can change over time:

- *Expenses can change, even if sales remain constant.* For example, you may expect an expense to decrease relative to sales over time if the firm will drive down costs by creating operating efficiencies or new production technologies. Alternately, you might expect an expense to increase relative to sales if the firm will face increasing input costs that it cannot pass along to customers by raising prices.
- *Sales can change, even if expenses remain constant.* For example, you may expect that the firm will hold expenses (such as cost of goods sold) relatively steady but will face increased competition for market share and therefore may be forced to lower sales prices, causing the expected expense-to-sales ratio to increase. Alternately, you might expect that sales will increase, but because of fixed costs in the production process, expenses will remain fairly steady.

- *Sales and expenses can change simultaneously and in the same direction.* If you expect both sales and operating expenses to increase (or decrease) simultaneously, the net result on the projected expense-to-sales percentage will depend on which of the two effects is proportionally greater.
- *Sales and expenses can change simultaneously but in opposite directions.* You might expect sales to increase while operating expenses decrease, which can occur for a firm in transition from the start-up phase to the growth phase of its life cycle. Or you might expect sales will decrease while operating expenses increase, as might occur for a firm in distress. The net result on the projected expense-to-sales percentage will depend on the relative magnitudes of the two effects.

In projecting the future relations between revenues and expenses, it is essential to evaluate the firm's strategies with respect to future growth, shifts in product/portfolio mix, the mix of fixed and variable expenses, competitive pressure on pricing, and many other factors that will impact expected future revenues and expenses.

Projecting Cost of Sales Including Occupancy Costs

Cost of sales and occupancy costs are the largest expenses on **Starbucks'** income statement. These expenses represent the costs of coffee beans and other products that **Starbucks** sells, as well as the occupancy costs (primarily rent) on company-operated stores. As such, these expenses represent a good example of expenses that contain fixed and variable components, as discussed in the preceding section. The rent component is largely fixed for a given company-operated store. The total amount of rent expense grows as **Starbucks** opens new company-operated stores. By contrast, the cost of sales component is likely to be more variable, varying with the amount of sales. We will therefore forecast each of the two components separately to capture the effects of these different drivers, and then we will combine them into the projected total amounts of cost of sales and occupancy costs.

The common-size income statement data discussed in Chapters 1 and 4 (and presented in the Analysis spreadsheet of FSAP) indicate that **Starbucks'** cost of goods sold as a percent of sales has steadily decreased from 42.9% in 2013 to 40.6% in 2015. However, the occupancy cost component has not been decreasing because **Starbucks** has been opening new company-operated stores. **Starbucks** discloses the total amount of annual rent expenses in Note 10, "Leases" (Appendix A). In 2015, for example, **Starbucks** incurred \$1,137.8 million in rent expense, up considerably from \$974.2 million in 2014. **Starbucks** had 11,474 company-operated stores on average during 2015 [(10,713 stores at the beginning of 2015 + 12,235 stores at the end of 2015)/2]. Therefore, **Starbucks** incurred \$0.099 million in rent expense per average store during 2015.

We will project that **Starbucks'** average rent expense per store will increase by 3% due to general inflation in Year +1, so the average rent expense will be \$0.102 million. In our revenue forecasts, we projected that **Starbucks** will have 12,885 company-operated stores by the end of Year +1. Thus, **Starbucks** will operate an average of 12,560 stores during Year +1 [(12,235 stores at the beginning + 12,885 stores at the end)/2]. Therefore, we project **Starbucks** will incur a total of \$1,282.9 million in rent expense during Year +1 [\$0.102 million in rent expense per average store × 12,560 store-years]. We use similar computations to project rent expense amounts for Years +2 through +5.

Starbucks' cost of sales (net of the rent expense component) as a percentage of total revenues has been decreasing steadily, from 36.9% in 2013, to 35.8% in 2014, to 34.7%

in 2015. Clearly, **Starbucks** generates cost efficiencies in its production processes. We project that **Starbucks** will continue to improve its performance and reduce the cost of sales in the future. We project that cost of sales (net of rent expense) as a percentage of total revenues will continue to decline to 34.0% in Year +1, and then continue to decline gradually in Years +2 through +5. Therefore, in Year +1, we project **Starbucks'** cost of sales will total $7,370.6 million [$21,678.3 million × 34.0%].

Combining the two components, we project **Starbucks'** Year +1 cost of sales and occupancy costs will total $8,653.5 million [$1,282.9 million in rent expense + $7,370.6 million in cost of sales]. The cost of sales and occupancy cost forecasts through Year +5 are as follows:

	2013	2014	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Cost of sales and occupancy costs	$6,382.3	$6,858.8	$7,787.5	$8,653.5	$9,390.4	$10,336.7	$11,358.7	$12,461.9
As a percent of total revenues	42.9%	41.7%	40.6%	39.9%	39.7%	39.3%	39.0%	38.6%
Rent expense (Note 10: Leases)	$ 894.7	$ 974.2	$1,137.8	$1,282.9	$1,389.7	$ 1,501.8	$ 1,619.4	$ 1,742.8
Number of company-operated stores at fiscal year-end	10,143	10,713	12,235	12,885	13,535	14,185	14,835	15,485
Average number of company-operated stores		10,428	11,474	12,560	13,210	13,860	14,510	15,160
Average annual rent per company-operated store		$ 0.093	$ 0.099	$ 0.102	$ 0.105	$ 0.108	$ 0.112	$ 0.115
Expected inflation in annual rent				3.0%	3.0%	3.0%	3.0%	3.0%
Cost of sales (net of rent expense)	$5,487.6	$5,884.6	$6,649.7	$7,370.6	$8,000.6	$ 8,834.8	$ 9,739.2	$10,719.2
As a percent of total revenues	36.9%	35.8%	34.7%	34.0%	33.8%	33.6%	33.4%	33.2%

Projecting Store Operating Expenses and Other Operating Expenses

Starbucks' second-most-significant operating expense is store operating expenses, which primarily involve payroll and other costs to manage **Starbucks'** company-operated stores. These expenses tend to vary with the number of company-operated stores. In the MD&A section of the 2015 Annual Report, **Starbucks** even discloses these expenses expressed as a percentage of company-operated store revenues. Store operating expenses have been declining steadily as a percentage of company-operated store revenues, from 36.3% in 2013, to 35.7% in 2014, to 35.6% in 2015. We project that these expenses will continue to decline as a percentage of store-operated revenues, from 35.0% in Year +1 to 34.2% in Year +5.

Starbucks' other operating expenses involve payroll, marketing, and various other costs to manage the licensing, CPG, and foodservice businesses. These expenses seem to vary with the total revenues from licensing, CPG, and Foodservice. In 2014 and 2015, these expenses amounted to 13.2% of those revenues. We will project that other operating expenses will decline slightly to 13.0% of total revenues from licensing, CPG, and

Foodservice in Years +1 through +5. The projected store operating expenses and other operating expenses through Year +5 are as follows:

Store Operating Expenses and Other Operating Expenses	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Company-operated store revenues	$15,203.2	$17,135.0	$18,671.3	$20,701.8	$22,917.5	$25,334.0
Store operating expenses	$ 5,411.1	$ 5,997.2	$ 6,497.6	$ 7,162.8	$ 7,883.6	$ 8,664.2
As a percent of company-operated store revenues	35.6%	35.0%	34.8%	34.6%	34.4%	34.2%
Licensing revenues	$ 1,856.0	$ 2,185.0	$ 2,454.0	$ 2,792.6	$ 3,162.1	$ 3,565.0
CPG, Foodservice, and Other revenues	$ 2,103.5	$ 2,358.3	$ 2,545.2	$ 2,799.8	$ 3,079.7	$ 3,387.7
Totals	$ 3,959.5	$ 4,543.4	$ 4,999.2	$ 5,592.4	$ 6,241.8	$ 6,952.7
Other operating expenses	$ 522.4	$ 590.6	$ 649.9	$ 727.0	$ 811.4	$ 903.9
As a percent of associated revenues	13.2%	13.0%	13.0%	13.0%	13.0%	13.0%

Projecting Property, Plant, and Equipment and Depreciation Expense

For a firm like **Starbucks**, with operating activities that are particularly dependent on property, plant, and equipment, you should create a separate schedule to forecast capital expenditures that lead projected future sales, and to forecast depreciation expense amounts that lag capital expenditures on property, plant, and equipment (PP&E).

In **Starbucks'** 2015 Annual Report, the MD&A section titled "Fiscal 2016—The View Ahead" discloses information about its expectations for the future.[9] In these disclosures, **Starbucks** states that it expects capital expenditures to be $1,400 million during fiscal 2016 (Year +1), primarily for new stores, store renovations, and other investments to support growth initiatives. This projected amount of capital expenditures is roughly 7.3% larger than capital expenditures in 2015, which amounted to $1,303.7 million. In prior years, **Starbucks'** management forecasts of capital expenditures have proven to be reliable, so we will use the projected $1,400 million for our Year +1 forecast.

In Years +2 through +5, we will assume **Starbucks'** capital expenditures will increase considerably. In developing our revenue forecasts, we projected that **Starbucks** will continue to open roughly the same number of new company-operated stores, so capital expenditures for new stores will likely remain fairly stable, growing with inflation. However, we believe **Starbucks** will have to increase capital spending for renovations and replacement of PP&E that is approaching the end of its useful life. At the end of fiscal 2015, **Starbucks** has 12,235 company-operated stores, but roughly 75% of those stores are at least five years old (some much older than that). Indeed, nearly 60% of **Starbucks'** existing PP&E has already been depreciated. Therefore, we project that **Starbucks** will incur capital expenditures of $1,500 million in Year +2; $1,800 million in Year +3; $2,100 million in Year +4; and $2,400 million in Year +5. We will add these capital expenditures to the PP&E totals on the projected balance sheets each year, and subtract these amounts of cash outflows in the investing section of the projected statements of cash flows each year.

For the income statement projections, we need to forecast depreciation expense each year. **Starbucks'** existing PP&E will continue to depreciate as it uses these assets in daily operations. In addition, **Starbucks'** capital expenditures on new PP&E will trigger a new layer of depreciation expense each year. **Starbucks** discloses in Note 1, "Summary of Significant Accounting Policies" (Appendix A), that it uses the straight-line depreciation method over the estimated useful lives of the assets. Note 1 does not disclose information related to salvage values, but we can assume that **Starbucks** depreciates PP&E to zero salvage value. Based on this assumption, we estimate the average useful life that **Starbucks** uses for depreciation by taking the average amount in PP&E at acquisition cost and dividing it by depreciation expense for that year. **Starbucks** discloses the gross amounts of PP&E at cost in Note 7, "Supplemental Balance Sheet Information." In fiscal 2015, for depreciation purposes **Starbucks** used an average useful life of 10.3 years {[($9,641.8 + $8,581.1)/2]/$883.8}. For simplicity, we will assume that **Starbucks** will use a 10-year average useful life for depreciation.

In computing depreciation expense for Year +1, we need to forecast two components. The first component is depreciation on the $9,641.8 million of existing PP&E as of the beginning of Year +1, which will be $964.2 million ($9,641.8/10.0 years). The second component is depreciation on the Year +1 capital expenditures of $1,400 million, which will be $140 million ($1,400/10.0 years). Together, total depreciation expense in Year +1 will be $1,104.2 million ($964.2 + $140.0), and accumulated depreciation (the PP&E contra account on the balance sheet) will grow by this amount of depreciation. In Year +2, depreciation expense will be $1,254.2 million, which consists of those two components plus a third component to reflect depreciation expense of $150.0 million ($1,500.0/10.0 years) on Year + 2 capital expenditures on PP&E, and so on. The projected amounts for capital expenditures and depreciation expense through Year +5 are as follows:

Capital Expenditures and Depreciation Expense Forecasts

	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Capital expenditures	$1,303.7	$ 1,400.0	$ 1,500.0	$ 1,800.0	$ 2,100.0	$ 2,400.0
Depreciation Expense Forecast Development		**Depreciation expense forecast on existing PP&E**				
Existing PP&E at cost	$9,641.8	$ 964.2	$ 964.2	$ 964.2	$ 964.2	$ 231.6
Remaining balance to be depreciated	$4,088.3	$ 3,124.1	$ 2,159.9	$ 1,195.8	$ 231.6	$ 0.0
Depreciation on New PP&E		**Depreciation expense forecasts on new PP&E**				
Capital expenditures Year +1	$1,400.0	$ 140.0	$ 140.0	$ 140.0	$ 140.0	$ 140.0
Capital expenditures Year +2	$1,500.0		$ 150.0	$ 150.0	$ 150.0	$ 150.0
Capital expenditures Year +3	$1,800.0			$ 180.0	$ 180.0	$ 180.0
Capital expenditures Year +4	$2,100.0				$ 210.0	$ 210.0
Capital expenditures Year +5	$2,400.0					$ 240.0
Total depreciation expense		**$1,104.2**	**$1,254.2**	**$1,434.2**	**$1,644.2**	**$1,151.6**
		Portion Reported Separately on Income Statement (95% of total)				
		$ 1,049.0	$ 1,191.5	$ 1,362.5	$ 1,562.0	$ 1,094.0
		Portion Reported within Cost of Sales on Income Statement (5% of total)				
		$ 55.2	$ 62.7	$ 71.7	$ 82.2	$ 57.6

Starbucks reports depreciation expense as a separate line item on the income statement, but it allocates a portion of depreciation expense to cost of sales, for depreciation on assets used in production and distribution of products. In 2015, for example, **Starbucks** added $933.8 million in depreciation and amortization expense back to net income on the statement of cash flows, but it reported only $893.9 million of depreciation and amortization expense on the income statement, which is roughly 95% of the total. The remaining 5% of the depreciation expense was included in cost of sales and occupancy costs. Therefore, in our income statement projections, we include 95% of the projected depreciation expense amounts as a separate line item for depreciation and amortization expense, and we implicitly assume the remaining 5% is included in our projections of cost of sales and occupancy. However, we do add the entire amount of depreciation and amortization expense back to net income in our projected statement of cash flows, discussed in a later section of this chapter.

On our projected balance sheets, we add these projected amounts for capital expenditures and depreciation expenses to PP&E, and accumulated depreciation, as follows (allow for rounding):

Property, Plant and Equipment and Accumulated Depreciation Forecasts

PP&E at cost	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Beginning balance at cost		$ 9,641.8	$11,041.8	$12,541.8	$ 14,341.8	$ 16,441.8
Add: CAPEX forecasts		1,400.0	1,500.0	1,800.0	2,100.0	2,400.0
Ending balance at cost	$ 9,641.8	$11,041.8	$12,541.8	$14,341.8	$ 16,441.8	$ 18,841.8
Accumulated depreciation						
Beginning balance		$(5,553.5)	$(6,657.7)	$(7,911.9)	$ (9,346.0)	$(10,990.2)
Less: Depreciation expense forecasts		(1,104.2)	(1,254.2)	(1,434.2)	(1,644.2)	(1,151.6)
Ending balance	$(5,553.5)	$(6,657.7)	$(7,911.9)	$(9,346.0)	$(10,990.2)	$(12,141.8)
PP&E—net	$ 4,088.3	$ 4,384.1	$ 4,629.9	$ 4,995.8	$ 5,451.6	$ 6,700.0

When you forecast fixed assets for capital-intensive firms (such as manufacturing firms or utility companies) or firms for which fixed-asset growth is a critical driver of future sales growth and earnings (for example, new stores for retail chains or restaurant chains), PP&E will typically be a large proportion of total assets and have a material impact on your forecasts. For such firms, you should invest considerable time and effort in developing detailed forecasts of capital expenditures, PP&E, and depreciation expense.[10] In FSAP, the Forecast Development spreadsheet includes a model for forecasting capital expenditures, PP&E, depreciation expense, and accumulated depreciation. The FSAP output (Appendix C) demonstrates the use of this model to compute the preceding forecasts for **Starbucks**.

FSAP

Projecting General and Administrative Expenses

The common-size income statement data reveal that **Starbucks'** general and administrative expenses are a steady percentage of total revenues, varying in a narrow range from 6.3% of total revenues in 2013, to 6.0% in 2014, to 6.2% in 2015. Going forward, we project general and administrative expenses to be roughly 6.0% of total revenues in the future. The projected amounts through Year +5 are as follows:

	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Total net revenues	$19,162.7	$21,678.3	$23,670.5	$26,294.2	$29,159.4	$32,286.7
General and administrative expenses	$ 1,196.7	$ 1,300.7	$ 1,420.2	$ 1,577.7	$ 1,749.6	$ 1,937.2
As a percent of total revenues	6.2%	6.0%	6.0%	6.0%	6.0%	6.0%

Projecting Income from Equity Investees

As Chapter 8 describes, some firms make substantial investments in affiliated companies or joint ventures that enable the investor company to exert significant influence but not control over the operating and financing decisions of the investee. In such cases, the investor might own 20% to 50% of the outstanding shares of the affiliate company but does not own enough of the shares to control the firm's activities, and therefore does not report the investment on a consolidated basis. Instead, under U.S. GAAP and IFRS, the investor company reports in income its proportionate share of the net income of the affiliate. On the balance sheet, equity investees (also called investments in noncontrolled affiliates) should increase with the firm's proportionate share of the net income of the affiliate, minus dividends received from the affiliate each period, plus or minus any additional investments or dispositions of investments in such affiliates.

To forecast future equity income from these types of equity investments, you can project a normal rate of return and the level of investment in equity affiliates. Alternately, a more time-consuming but potentially more accurate approach would be to prepare a full set of financial statement forecasts for the equity affiliates and estimate the investor's share of expected future income. This approach is worthwhile for firms that have very large and important investments in joint ventures and affiliated companies.

Starbucks includes income from equity investees in operating income. It likely makes this reporting choice because many of **Starbucks'** equity investments involve joint ventures or affiliated firms with operating activities that relate closely to **Starbucks'** operating activities. For example, prior to 2015, one of the joint ventures was **Starbucks** Japan, in which **Starbucks** owned a large but noncontrolling proportion (39.5%) of the shares of the firm that licensed and operated 1,009 **Starbucks** coffee shops in Japan. In fiscal 2015, **Starbucks** acquired 100% ownership of the firm, so its financial statements are now consolidated with **Starbucks'** financial statements, and **Starbucks** now reports the stores as company-operated stores. **Starbucks** discloses in Note 6, "Equity and Cost Investments," that at the end of fiscal 2015, equity method investments include roughly 50% ownership interests in joint ventures operating licensed stores in China, Korea, Taiwan, India, Spain, and others, as well as 50% ownership in The North American Coffee Partnership with PepsiCo.

Equity investments generate a substantial amount of operating income for **Starbucks**. In 2015, **Starbucks** included $249.9 million in income from equity investees in operating income. As it describes in Note 6, this amount includes **Starbucks'** proportionate share of the income (and losses) from these equity investees. In addition, **Starbucks** includes its share of gross profits from selling products to these equity investees, as well as license fees and royalties. Dividing the amount of income from equity investees by the average balance of the investments in equity investees reveals that **Starbucks** has generated returns in excess of 50% on these investments over the past three years. In 2015, for example, **Starbucks** generated a return of 57.7% ($249.9 million in income divided by an average investment balance of $433.5 million [($514.9 million at the beginning of the year + $352.0 million at the end)/2]. Going forward, we will assume **Starbucks** will continue to generate a 50% return on the average amount invested in equity investees each year. We will also assume that these investments will grow by roughly 5% per year over the five-year forecast horizon.[11] The projected investment amounts and income from equity investees are as follows (allow for rounding):

Investments in and Income from Equity Investees

Equity Investees	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Beginning balance	$352.0	$369.6	$388.1	$407.5	$427.9	$449.3
Assume 5% growth		5.0%	5.0%	5.0%	5.0%	5.0%
Income from equity investees						
Average investment balance		$360.8	$378.8	$397.8	$417.7	$438.6
Projected rate of return		50.0%	50.0%	50.0%	50.0%	50.0%
Projected income		$180.4	$189.4	$198.9	$208.8	$219.3

Projecting Nonrecurring Income Items

As discussed in prior chapters, it is not uncommon for firms' reported income statements to include other nonrecurring gains or losses that are part of operations, unusual gains or losses that are peripheral to operations, and income from discontinued segments. For example, in the 2015 income statement, **Starbucks** recognized a $390.6 million gain resulting from an acquisition of a joint venture, as well as a $61.1 million loss on the early extinguishment of long-term debt. In fiscal 2013, **Starbucks** recognized a huge $2,784.1 million litigation charge for a legal settlement with Kraft.

As previous chapters discussed, you must determine whether items such as these are likely to persist in the future; if so, include them in the financial statement forecasts.[12] It is very difficult to predict specific future amounts of nonrecurring items for **Starbucks** because they seem to be infrequent and nonrecurring. Therefore, we will assume that **Starbucks** will not generate any nonrecurring gains or losses in income over our forecast horizon.

Projecting Operating Income

The revenue forecasts—together with the projected costs of sales and occupancy costs, store operating expenses, other operating expenses, depreciation and amortization expenses, general and administrative expenses, and income from equity investees,

[11]For the analyst who needs greater forecast precision, the projections of future balances in Equity Investments in Noncontrolled Affiliates should follow the accounting methods for equity method investments. As such, the balances should grow with the firm's proportionate share of the net income of the affiliate minus dividends received from the affiliate each period, and should increase with additional investments and decrease with dispositions of investments in such affiliates.

SG&A expenses, and amortization of intangible assets—lead to the following projected amounts of operating income for **Starbucks** for Years +1 through +5:

Operating Income Projections	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Total net revenues	**$19,162.7**	**$21,678.3**	**$23,670.5**	**$ 26,294.2**	**$ 29,159.4**	**$ 32,286.7**
Cost of sales and occupancy	(7,787.5)	(8,653.5)	(9,390.4)	(10,336.7)	(11,358.7)	(12,461.9)
Store operating expenses	(5,411.1)	(5,997.2)	(6,497.6)	(7,162.8)	(7,883.6)	(8,664.2)
Other operating expenses	(522.4)	(590.6)	(649.9)	(727.0)	(811.4)	(903.9)
Depreciation expense	(893.9)	(1,049.0)	(1,191.5)	(1,362.5)	(1,562.0)	(1,094.0)
General and administrative expenses	(1,196.7)	(1,300.7)	(1,420.2)	(1,577.7)	(1,749.6)	(1,937.2)
Income from equity investees	249.9	180.4	189.4	198.9	208.8	219.3
Operating income	**$ 3,601.0**	**$ 4,267.7**	**$ 4,710.4**	**$ 5,326.4**	**$ 6,002.9**	**$ 7,444.8**
Operating income margin	18.8%	19.7%	19.9%	20.3%	20.6%	23.1%

Exhibit 10.3 presents the complete forecasts of **Starbucks'** income statements, as well as comprehensive income, for Years +1 through +5. The exhibit also presents forecast amounts for Year +6, assuming **Starbucks** will grow at a constant rate of 3%. The format of this exhibit mirrors the format of the Forecasts spreadsheet in FSAP. In later sections of this chapter, we discuss the projections of interest income, interest expense, income tax expense, net income, comprehensive income, and the change in retained earnings, after projecting **Starbucks'** balance sheet.

LO 10-2c

Build forecasts of future balance sheets, income statements, and statements of cash flows by applying the seven-step forecasting framework to project: c. operating assets and liabilities

Step 3: Project Operating Assets and Liabilities on the Balance Sheet

In this section, we describe how the operating activities we projected for the income statement will give rise to future operating assets and liabilities on the balance sheet. We demonstrate how to forecast balance sheet amounts using various drivers of growth in different assets and liabilities.

Techniques to Project Operating Assets and Liabilities

To forecast individual operating assets and liabilities, you must first determine the underlying operating activities that drive them. For some types of assets, such as inventory and property, plant, and equipment, asset growth typically *leads* future sales growth. Growth for other types of assets, such as accounts receivable, typically *lags* sales growth. Certain operating liabilities will be determined by operating assets (such as accounts payable arising from inventory purchases), whereas others will be determined by operating expenses (such as accrued expenses).

After determining the business activities that drive future operating assets and liabilities, you can adopt a number of techniques to forecast the future balance sheet amounts. You might project certain types of assets and liabilities using a growth rate, such as the expected growth in sales, for assets and liabilities that vary with sales, such as future accounts receivable. For some types of assets and liabilities, you might project future growth rates based on past growth trends or expected shifts in

Exhibit 10.3

Starbucks
Actual and Forecasted Statements of Net Income and Comprehensive Income
(amounts in millions; allow for rounding)

Actual and forecasted amounts in bold; below the actual amounts (only) we report historical common-size and rate-of-change percentages. Below the forecast amounts (only) we report the forecast assumptions and brief explanations.

	Actuals			Forecasts					
	2013	2014	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6
INCOME STATEMENT									
Revenues	**$14,866.8**	**$16,447.8**	**$19,162.7**	**$21,678.3**	**$23,670.5**	**$ 26,294.2**	**$ 29,159.4**	**$ 32,286.7**	**$ 33,255.3**
common size	100.0%	100.0%	100.0%	13.1%	9.2%	11.1%	10.9%	10.7%	
rate of change		10.6%	16.5%	See Forecast Development worksheet for details of revenues forecasts.					
Cost of sales and occupancy expense	**(6,382.3)**	**(6,858.8)**	**(7,787.5)**	**(8,653.5)**	**(9,390.4)**	**(10,336.7)**	**(11,358.7)**	**(12,461.9)**	**(12,835.8)**
common size	(42.9%)	(41.7%)	(40.6%)	(39.9%)	(39.7%)	(39.3%)	(39.0%)	(38.6%)	
rate of change		7.5%	13.5%	See Forecast Development worksheet for details.					
Gross profit	**$ 8,484.5**	**$ 9,589.0**	**$11,375.2**	**$13,024.9**	**$14,280.2**	**$ 15,957.5**	**$ 17,800.7**	**$ 19,824.8**	**$ 20,419.5**
common size	57.1%	58.3%	59.4%	60.1%	60.3%	60.7%	61.0%	61.4%	61.4%
rate of change		13.0%	18.6%	14.5%	9.6%	11.7%	11.6%	11.4%	
Store operating expenses	**(4,286.1)**	**(4,638.2)**	**(5,411.1)**	**(5,997.2)**	**(6,497.6)**	**(7,162.8)**	**(7,883.6)**	**(8,664.2)**	**(8,924.2)**
common size	(28.8%)	(28.2%)	(28.2%)	(35.0%)	(34.8%)	(34.6%)	(34.4%)	(34.2%)	
rate of change		8.2%	16.7%	Assume store operating expenses decline as a percent of company-operated store revenues.					
Other operating expenses	**(431.8)**	**(457.3)**	**(522.4)**	**(590.6)**	**(649.9)**	**(727.0)**	**(811.4)**	**(903.9)**	**(931.0)**
common size	(2.9%)	(2.8%)	(2.7%)	(13.0%)	(13.0%)	(13.0%)	(13.0%)	(13.0%)	
rate of change		5.9%	14.2%	Assume steady percentage of licensed, CPG, Foodservice, and other revenues.					
Depreciation and amortization	**(621.4)**	**(709.6)**	**(893.9)**	**(1,049.0)**	**(1,191.5)**	**(1,362.5)**	**(1,562.0)**	**(1,094.0)**	**(1,126.8)**
common size	(4.2%)	(4.3%)	(4.7%)						
rate of change		14.2%	26.0%	Amounts from depreciation schedule, Forecast Development worksheet.					
Gen. and admin. expenses	**(937.9)**	**(991.3)**	**(1,196.7)**	**(1,300.7)**	**(1,420.2)**	**(1,577.7)**	**(1,749.6)**	**(1,937.2)**	**(1,995.3)**
common size	(6.3%)	(6.0%)	(6.2%)	(6.0%)	(6.0%)	(6.0%)	(6.0%)	(6.0%)	
rate of change		5.7%	20.7%	Assume steady state relative to revenues.					
Income from equity investees	**251.4**	**268.3**	**249.9**	**180.4**	**189.4**	**198.9**	**208.8**	**219.3**	**225.9**
common size	1.7%	1.6%	1.3%	50.0%	50.0%	50.0%	50.0%	50.0%	
rate of change		6.7%	(6.9%)	Assume 50% rate of return from equity investees.					

(Continued)

Exhibit 10.3 (Continued)

	Actuals			Forecasts					
	2013	2014	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6
Nonrecurring operating gains losses	(2,784.1)	20.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0
common size	(18.7%)	0.1%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	
rate of change		(100.7%)	(100.0%)	Assume zero nonrecurring items.					
Operating profit	$ (325.4)	$ 3,081.1	$ 3,601.0	$ 4,267.7	$ 4,710.4	$ 5,326.4	$ 6,002.9	$ 7,444.8	$ 7,668.1
common size	(2.2%)	18.7%	18.8%	19.7%	19.9%	20.3%	20.6%	23.1%	23.1%
rate of change		(1,046.9%)	16.9%	18.5%	10.4%	13.1%	12.7%	24.0%	
Interest income	123.6	142.7	43.0	43.2	47.8	52.0	57.0	62.5	64.4
common size	0.8%	0.9%	0.2%	2.1%	2.1%	2.1%	2.1%	2.1%	
rate of change		15.5%	(69.9%)	Interest rate earned on average balance in cash and investment securities.					
Interest expense	(28.1)	(64.1)	(70.5)	(77.1)	(90.6)	(90.6)	(90.6)	(90.6)	(93.3)
common size	(0.2%)	(0.4%)	(0.4%)	(2.71%)	(2.71%)	(2.71%)	(2.71%)	(2.71%)	
rate of change		128.1%	10.0%	Weighted-average interest rate on average balances in long-term debt. See Forecast Development.					
Income before Tax	$ (229.9)	$ 3,159.7	$ 3,903.0	$ 4,233.8	$ 4,667.5	$ 5,287.8	$ 5,969.3	$ 7,416.6	$ 7,639.1
common size	(1.5%)	19.2%	20.4%	19.5%	19.7%	20.1%	20.5%	23.0%	23.0%
rate of change		(1474.4%)	23.5%	8.5%	10.2%	13.3%	12.9%	24.2%	
Income tax expense	238.7	(1,092.0)	(1,143.7)	(1,439.5)	(1,587.0)	(1,797.9)	(2,029.6)	(2,521.7)	(2,597.3)
common size	1.6%	(6.6%)	(6.0%)	(34.0%)	(34.0%)	(34.0%)	(34.0%)	(34.0%)	
rate of change		(557.5%)	4.7%	Effective income tax rate assumptions.					
Net income	$ 8.8	$ 2,067.7	$ 2,759.3	$ 2,794.3	$ 3,080.6	$ 3,490.0	$ 3,939.8	$ 4,895.0	$ 5,041.8
common size	0.1%	12.6%	14.4%	12.9%	13.0%	13.3%	13.5%	15.2%	15.2%
rate of change		23,396.6%	33.4%	1.3%	10.2%	13.3%	12.9%	24.2%	3.0%
Net income attributable to noncontrolling interests	(0.5)	0.4	(1.9)	0.0	0.0	0.0	0.0	0.0	0.0
common size	0.0%	0.0%	0.0%	0.0	0.0	0.0	0.0	0.0	
rate of change		(180.0%)	(575.0%)	Assume noncontrolling interests are acquired.					
Net income attributable to common shareholders	$ 8.3	$ 2,068.1	$ 2,757.4	$ 2,794.3	$ 3,080.6	$ 3,490.0	$ 3,939.8	$ 4,895.0	$ 5,041.8
common size	0.1%	12.6%	14.4%	12.9%	13.0%	13.3%	13.5%	15.2%	15.2%
rate of change		24,816.9%	33.3%	1.3%	10.2%	13.3%	12.9%	24.2%	3.0%
Other comprehensive income items	43.8	(41.3)	(195.5)	0.0	0.0	0.0	0.0	0.0	0.0
common size	0.3%	(0.3%)	(1.0%)	0.0	0.0	0.0	0.0	0.0	
rate of change		(194.3%)	373.4%	Assume random walk, mean zero.					
Comprehensive income	$ 52.6	$ 2,026.4	$ 2,563.8	$ 2,794.3	$ 3,080.6	$ 3,490.0	$ 3,939.8	$ 4,895.0	$ 5,041.8
common size	0.4%	12.3%	13.4%	12.9%	13.0%	13.3%	13.5%	15.2%	15.2%
rate of change		3,752.5%	26.5%	9.0%	10.2%	13.3%	12.9%	24.2%	3.0%

the firm's strategy. For example, perhaps the firm plans to hold larger amounts of cash than it has held in the past in order to increase its liquidity. In other cases, the firm may have a strategy of maintaining certain types of assets and liabilities at relatively steady proportions of total assets, in which case you might project future amounts using the expected future common-size percentages. This approach would work, for example, if the firm maintains a target percentage of total assets in cash. In some instances, asset and liability amounts can be projected based on large future transactions. For instance, if a firm is expected to borrow a large amount of cash in order to construct a new plant, you might project a large increase in cash after the borrowing and a large decrease following the investment.

You might choose to project future amounts for operating assets and liabilities, such as cash; accounts receivable; inventory; property, plant, and equipment; and accounts payable using the turnover rates demonstrated in Chapter 4. Using turnover rates produces reasonable forecasts of average and year-end account balances if the firm generates revenues evenly throughout the year and if the forecasted account varies reliably with revenues. However, you should not use a turnover-based forecast if the firm will experience substantially different future growth rates in revenues and the forecasted account, or if the turnover rate varies unpredictably over time.

A less desirable feature that can result from using a sales-turnover forecasting approach is that if the firm has exhibited volatility in historical amounts, it can trigger volatility in forecast amounts as well. To illustrate, suppose you expect that sales will remain constant at \$12,167 per year over the next five years. Suppose also that you expect the firm will hold roughly 30 days of sales in cash and that it currently has a cash balance of \$800. Using the cash turnover rate of 30 days' sales, you would project the firm will maintain an average cash balance of \$1,000 (\$12,167/365 $\times$ 30).

To compute the year-end cash balance from the average cash balance, you multiply the average by two and then subtract the beginning balance [Ending = (Average $\times$ 2) − Beginning]. Applying this approach, the projected year-end cash balances would be as follows:

	Annual Sales Forecasts	Average Sales per Day	Days' Sales in Cash	Average Cash Balance	Beginning Cash Balance	Ending Cash Balance
Year +1	\$12,167	\$33.33	30	\$1,000	\$ 800	\$1,200
Year +2	\$12,167	\$33.33	30	\$1,000	\$1,200	\$ 800
Year +3	\$12,167	\$33.33	30	\$1,000	\$ 800	\$1,200
Year +4	\$12,167	\$33.33	30	\$1,000	\$1,200	\$ 800
Year +5	\$12,167	\$33.33	30	\$1,000	\$ 800	\$1,200

Notice that, because the firm held a smaller-than-average cash balance at the start of the forecast period (\$800), the forecast of ending cash will have to compensate with a larger-than-average cash balance (\$1,200) at the end of Year +1. The forecast model will then have to compensate again in Year +2 and project a smaller-than-average year-end balance (\$800). The relatively small balance at the beginning of Year +1 triggers a relatively large balance at the end of Year +1, which in turn triggers a relatively small balance at the end of Year +2, and so on, creating a sawtooth pattern of variability.

For some firms, this type of variability is a realistic outcome of seasonal or cyclical volatility in the firm's operating activities. Forecasts that capture seasonality or cyclicality are preferable when you are concerned about whether a firm might violate certain contractual constraints, such as debt covenants or regulatory capital requirements. In other contexts, you may prefer smooth forecasts that mitigate variability. Smooth forecasts are often preferable in contexts where you expect random fluctuations around a generally smooth average growth trend over time. Smooth growth forecasts also tend to be easier to present and explain to an audience.

A number of techniques exist for you to produce smooth forecasts. One such technique is to project the ending balances to equal the projected average balances. In the example, one would simply assume that the projected ending balance in cash each year will equal the average balance ($1,000). Another forecast smoothing technique involves using turnover rates to project the ending balance in the forecast account at the end of the forecast horizon (say, Year +5). Then forecast the annual amounts in Year +1, +2, and so on, using the steady growth rate necessary to reach the projected ending amount in Year +5. Alternately, if you expect the ending balance will vary with sales, then you can simply project growth using the expected sales growth rates. In choosing among forecasting techniques, you must trade off the objectives of achieving minimizing forecast error while avoiding unnecessary computational complexity.

In the sections that follow, we project individual operating assets and liabilities for **Starbucks** using a combination of forecast drivers, including common-size percentages, growth rates, and asset turnovers. We proceed in roughly the order in which **Starbucks** presents its accounts on the balance sheet. Exhibit 10.4 provides a preview of the projected balance sheets for **Starbucks** for Year +1 through Year +5, which we developed using the Forecasts spreadsheet in FSAP. The exhibit also presents forecast amounts for Year +6, assuming **Starbucks** will grow at a constant rate of 3%.

Projecting Cash and Cash Equivalents

The Analysis spreadsheet in FSAP computes the average turnover of cash through revenues each year, so we use that turnover rate to project **Starbucks'** ending cash balances. Like all firms, **Starbucks** needs a certain amount of cash on hand for day-to-day liquidity.[13] During 2015, **Starbucks** had average cash balances of roughly 30.8 days of sales (computed as 365 days divided by the ratio of revenues to the average balance in cash; in 2015, 30.8 days = 365/{$19,162.7/[($1,530.1 + $1,708.4)/2]}. We assume that **Starbucks** will maintain ending cash balances equivalent to 30 days of sales in the future.

To apply this approach, we use our forecasts of revenues and the projected number of days' sales in cash to compute the ending balance in cash each year. The Year +1 revenue forecast is $21,678.3 million, or an average of $59.4 million per day.[14] We project **Starbucks** will hold 30 days of sales in cash at the end of Year +1, for a cash

Exhibit 10.4

Starbucks
Actual and Forecasted Balance Sheets
(amounts in millions; allow for rounding)

The actual and forecasted amounts are in bold. Below the actual amounts (only), we report historical common-size and rate-of-change-percentages. Below the forecast amounts (only), we report the forecast assumptions and brief explanations.

	Actuals			Forecasts					
	2013	2014	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6
BALANCE SHEET									
ASSETS									
Cash and cash equivalents	**$ 2,575.7**	**$ 1,708.4**	**$ 1,530.1**	**$ 1,781.8**	**$ 1,945.5**	**$ 2,161.2**	**$ 2,396.7**	**$ 2,653.7**	**$ 2,733.3**
common size	22.4%	15.9%	12.3%	30.0	30.0	30.0	30.0	30.0	
rate of change		(33.7%)	(10.4%)	Assume ending cash balances equal to 30 days sales.					
Short-term investments	**658.1**	**135.4**	**81.3**	**83.7**	**86.3**	**88.8**	**91.5**	**94.2**	**97.1**
common size	5.7%	1.3%	0.7%	3.0%	3.0%	3.0%	3.0%	3.0%	
rate of change		(79.4%)	(40.0%)	Assume 3% growth.					
Accounts and notes receivable—net	**561.4**	**631.0**	**719.0**	**825.0**	**907.8**	**1,015.5**	**1,133.4**	**1,262.5**	**1,300.4**
common size	4.9%	5.9%	5.8%	14.7%	10.0%	11.9%	11.6%	11.4%	
rate of change		12.4%	13.9%	Assume growth with licensing and CPG revenues					
Inventories	**1,111.2**	**1,090.9**	**1,306.4**	**1,327.7**	**1,440.7**	**1,585.9**	**1,742.7**	**1,912.0**	**1,969.3**
common size	9.6%	10.1%	10.5%	56.0	56.0	56.0	56.0	56.0	
rate of change		(1.8%)	19.8%	Assume ending inventory equals 56 days' cost of goods sold.					
Prepaid expenses and other current assets	**287.7**	**285.6**	**334.2**	**352.0**	**369.7**	**387.5**	**405.2**	**423.0**	**435.7**
common size	2.5%	2.7%	2.7%	5.3%	5.0%	4.8%	4.6%	4.4%	
rate of change		(0.7%)	17.0%	Assume growth with company-operated stores.					
Deferred income taxes—current	**277.3**	**317.4**	**381.8**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**
common size	2.4%	3.0%	3.1%	0.0%	0.0%	0.0%	0.0%	0.0%	
rate of change		14.5%	20.3%	Assume the current portion of deferred tax assets is fully realized in Year +1.					
Current assets	**$ 5,471.4**	**$ 4,168.7**	**$ 4,352.8**	**$ 4,370.2**	**$ 4,750.0**	**$ 5,238.9**	**$ 5,769.5**	**$ 6,345.4**	**$ 6,535.8**
common size	47.5%	38.8%	35.0%	34.1%	35.1%	36.1%	37.0%	36.2%	36.2%
rate of change		(23.8%)	4.4%	0.4%	8.7%	10.3%	10.1%	10.0%	3.0%
Long-term investments	**58.3**	**318.4**	**312.5**	**321.9**	**331.5**	**341.5**	**351.7**	**362.3**	**373.1**
common size	0.5%	3.0%	2.5%	3.0%	3.0%	3.0%	3.0%	3.0%	
rate of change		446.1%	(1.9%)	Assume steady growth at 3%.					

Exhibit 10.4 (Continued)

	Actuals			Forecasts					
	2013	2014	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6
Equity and cost investments	496.5	514.9	352.0	369.6	388.1	407.5	427.9	449.3	462.7
common size	4.3%	4.8%	2.8%	5.0%	5.0%	5.0%	5.0%	5.0%	
rate of change		3.7%	(31.6%)	Assume steady growth at 5%.					
Property, plant, and equipment—at cost	7,782.1	8,581.1	9,641.8	11,041.8	12,541.8	14,341.8	16,441.8	18,841.8	19,407.1
common size	67.6%	79.8%	77.5%						
rate of change		10.3%	12.4%	PP&E assumptions—see schedule in forecast development.					
Accumulated depreciation	(4,581.6)	(5,062.1)	(5,553.5)	(6,657.7)	(7,911.9)	(9,346.0)	(10,990.2)	(12,141.8)	(12,506.1)
common size	(39.8%)	(47.1%)	(44.6%)						
rate of change		10.5%	9.7%	See depreciation schedule in forecast development worksheet.					
Deferred income taxes—noncurrent	967.0	903.3	828.9	746.0	671.4	604.3	543.8	489.5	504.1
common size	8.4%	8.4%	6.7%	(10.0%)	(10.0%)	(10.0%)	(10.0%)	(10.0%)	
rate of change		(6.6%)	(8.2%)	Assume noncurrent deferred tax assets decrease 10% per year, as tax benefits are realized.					
Other assets	185.3	198.9	415.9	436.7	458.5	481.5	505.5	530.8	546.7
common size	1.6%	1.8%	3.3%	5.0%	5.0%	5.0%	5.0%	5.0%	
rate of change		7.3%	109.1%	Assume 5% growth.					
Other intangible assets	274.8	273.5	520.4	546.4	573.7	602.4	632.5	664.2	684.1
common size	2.4%	2.5%	4.2%	5.0%	5.0%	5.0%	5.0%	5.0%	
rate of change		(0.5%)	90.3%	Assume 5% growth.					
Goodwill	862.9	856.2	1,575.4	1,654.2	1,736.9	1,823.7	1,914.9	2,010.7	2,071.0
common size	7.5%	8.0%	12.7%	5.0%	5.0%	5.0%	5.0%	5.0%	
rate of change		(0.8%)	84.0%	Assume 5% growth.					
Total Assets	$11,516.7	$10,752.9	$12,446.2	$12,829.0	$13,540.1	$14,495.5	$ 15,597.5	$ 17,552.1	$ 18,078.6
common size	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
rate of change		(6.6%)	15.7%	3.1%	5.5%	7.1%	7.6%	12.5%	3.0%
LIABILITIES									
Accounts payable	$ 491.7	$ 533.7	$ 684.2	$ 665.5	$ 729.0	$ 804.1	$ 883.4	$ 969.0	$ 998.0
common size	4.3%	5.0%	5.5%	28.0	28.0	28.0	28.0	28.0	
rate of change		8.5%	28.2%	Assume a 28-day payment period consistent with recent years.					
Accrued liabilities	1,269.3	1,514.4	1,760.7	1,991.8	2,174.9	2,416.0	2,679.2	2,966.6	3,055.6
common size	11.0%	14.1%	14.1%	13.1%	9.2%	11.1%	10.9%	10.7%	
rate of change		19.3%	16.3%	Assume growth with total revenues.					

	Actuals			Forecasts					
	2013	2014	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6
Current maturities of long-term debt	0.0	0.0	0.0	400.0	0.0	350.0	0.0	0.0	0.0
common size	0.0%	0.0%	0.0%	0.0	0.0	0.0	0.0	0.0	
rate of change				Current maturities of long-term debt per long-term debt note (Note 9).					
Insurance reserves	178.5	196.1	224.8	231.5	238.5	245.6	253.0	260.6	268.4
common size	1.5%	1.8%	1.8%	3.0%	3.0%	3.0%	3.0%	3.0%	
rate of change		9.9%	14.6%	Assume 3% growth.					
Stored value card liability	653.7	794.5	983.8	1,113.0	1,215.2	1,349.9	1,497.0	1,657.6	1,707.3
common size	5.7%	7.4%	7.9%	13.1%	9.2%	11.1%	10.9%	10.7%	
rate of change		21.5%	23.8%	Assume growth with total revenues.					
Accrued litigation charge	2,784.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
common size	24.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	
rate of change		(100.0%)		Assume nonrecurring.					
Current liabilities	$ 5,377.3	$ 3,038.7	$ 3,653.5	$ 4,401.8	$ 4,357.6	$ 5,165.6	$ 5,312.6	$ 5,853.7	$ 6,029.3
common size	46.7%	28.3%	29.4%	34.3%	32.2%	35.6%	34.1%	33.4%	33.4%
rate of change		(43.5%)	20.2%	20.5%	(1.0%)	18.5%	2.8%	10.2%	
Long-term debt	1,299.4	2,048.3	2,347.5	2,947.5	3,347.5	2,997.5	3,347.5	3,347.5	3,447.9
common size	11.3%	19.0%	18.9%	1,000.0	400.00	0.00	350.00	0.00	
rate of change		57.6%	14.6%	$1,000 new long-term debt issues in 2016; see debt maturities in Note 9.					
Long-term accrued liabilities	357.7	392.2	625.3	707.4	772.4	858.0	951.5	1,053.6	1,085.2
common size	3.1%	3.6%	5.0%	13.1%	9.2%	11.1%	10.9%	10.7%	
rate of change		9.6%	59.4%	Assume growth with G&A expenses, which grow with revenues.					
Total liabilities	$ 7,034.4	$ 5,479.2	$ 6,626.3	$ 8,056.7	$ 8,477.5	$ 9,021.1	$ 9,611.6	$ 10,254.8	$ 10,562.4
common size	61.1%	51.0%	53.2%	62.8%	62.6%	62.2%	61.6%	58.4%	58.4%
rate of change		(22.1%)	20.9%	21.6%	5.2%	6.4%	6.5%	6.7%	
SHAREHOLDERS' EQUITY									
Common stock + additional paid-in capital	282.9	40.1	42.6	43.9	46.3	49.6	53.4	60.1	61.9
common size	2.5%	0.4%	0.342%	0.342%	0.342%	0.342%	0.342%	0.342%	
rate of change		(85.8%)	6.2%	Assume steady percent of total assets.					
Retained earnings	4,130.3	5,206.6	5,974.8	4,927.8	5,215.6	5,624.1	6,131.9	7,436.6	7,653.7
common size	35.9%	48.4%	48.0%						
rate of change		26.1%	14.8%	Add net income and subtract dividends and share repurchases.					

Exhibit 10.4 (Continued)

	Actuals			Forecasts					
	2013	2014	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6
Accum. other comprehensive income <loss>	67.0	25.3	(199.4)	(199.4)	(199.4)	(199.4)	(199.4)	(199.4)	(199.4)
common size	0.6%	0.2%	(1.6%)	0.0	0.0	0.0	0.0	0.0	
rate of change		(62.2%)	(888.1%)	Add accumulated other comprehensive income items from income statement					
Total common shareholders' equity	$ 4,480.2	$ 5,272.0	$ 5,818.0	$ 4,772.4	$ 5,062.6	$ 5,474.4	$ 5,985.9	$ 7,297.3	$ 7,516.2
common size	38.9%	49.0%	46.7%	37.2%	37.4%	37.8%	38.4%	41.6%	
rate of change		17.7%	10.4%	(18.0%)	6.1%	8.1%	9.3%	21.9%	3.0%
Noncontrolling interests	2.1	1.7	1.8	0.0	0.0	0.0	0.0	0.0	0.0
common size	0.0%	0.0%	0.0%	0.0	0.0	0.0	0.0	0.0	
rate of change		(19.0%)	5.9%	Assume noncontrolling interests are acquired in Year +1.					
Total equity	$ 4,482.3	$ 5,273.7	$ 5,819.8	$ 4,772.4	$ 5,062.6	$ 5,474.4	$ 5,985.9	$ 7,297.3	$ 7,516.2
common size	38.9%	49.0%	46.8%	37.2%	37.4%	37.8%	38.4%	41.6%	
rate of change		17.7%	10.4%	(18.0%)	6.1%	8.1%	9.3%	21.9%	
Total liabilities and equities	$11,516.7	$10,752.9	$12,446.1	$12,829.0	$13,540.1	$14,495.5	$ 15,597.5	$ 17,552.1	$ 18,078.6
common size	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%	100.0%
rate of change		(6.6%)	15.7%	3.1%	5.5%	7.1%	7.6%	12.5%	3.0%

balance of $1,781.8 million. The projected balances in cash and cash equivalents on the projected **Starbucks** balance sheets follow (allow for rounding):

Cash and Cash Equivalents	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Projected revenues	$19,162.7	$21,678.3	$23,670.5	$26,294.2	$29,159.4	$32,286.7
Revenues per day	$ 52.5	$ 59.4	$ 64.9	$ 72.0	$ 79.9	$ 88.5
Projected cash turnover (in days)	30.8	30.0	30.0	30.0	30.0	30.0
Projected cash balances	$ 1,530.1	$ 1,781.8	$ 1,945.5	$ 2,161.2	$ 2,396.7	$ 2,653.7

For the three primary financial statement forecasts to articulate with each other, the change in the cash balance on the projected balance sheet each year must agree with the change in cash on the projected statement of cash flows. Later in the chapter, we demonstrate how to forecast the implied statement of cash flows.

Projecting Short-Term Investments

From 2013 to 2015, **Starbucks'** short-term investments balances (also known as marketable securities) dropped dramatically, from $658.1 million to $81.3 million. **Starbucks** sold some of the short-term investments in part to pay the large legal settlement with Kraft in 2014. It appears **Starbucks** primarily manages its liquidity through its fairly large cash balances rather than through its very small amount of short-term investments. Going forward, we will simply assume that **Starbucks'** short-term investments balances will grow by 3% per year. Later in this chapter, we will include on the forecasted income statements the future interest income that we expect the cash, short-term, and long-term investments to generate. The projected balances follow:

Short-Term Investments	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Projected balances	$81.3	$83.7	$86.3	$88.8	$91.5	$94.2
Projected growth rate		3.0%	3.0%	3.0%	3.0%	3.0%

Projecting Accounts Receivable

Starbucks' accounts receivable are driven by licensing, CPG, foodservice, and other revenues. **Starbucks'** makes sales to licensees, grocery store chains, and foodservice distributors on credit, which create accounts receivable. By contrast, **Starbucks** makes sales in company-operated stores either for cash, through the redemption of **Starbucks'** stored-value cards (gift cards, which are prepaid), or through third-party credit cards (e.g., Visa or Mastercard), which are reported as "cash equivalents" within the cash account. Company-operated store revenues do not create accounts receivable. We will therefore use the projected rate of growth in licensing, CPG, foodservice, and other revenues to project growth in accounts receivable. The projected amounts are as follows (allow for rounding):

Accounts Receivable	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Licensed store revenues	$1,856.0	$2,185.0	$2,454.0	$2,792.6	$3,162.1	$3,565.0
CPG, foodservice, and other revenues	$2,103.5	$2,358.3	$2,545.2	$2,799.8	$3,079.7	$3,387.7
Total	$3,959.5	$4,543.4	$4,999.2	$5,592.4	$6,241.8	$6,952.7
Growth rate		14.7%	10.0%	11.9%	11.6%	11.4%
Accounts receivable	$ 719.0	$ 825.0	$ 907.8	$1,015.5	$1,133.4	$1,262.5

Projecting Inventories

Chapter 4's analysis of **Starbucks'** inventory turnover ratios revealed that **Starbucks** has generated faster inventory turnover rates of 56 days in 2015, compared to 59 days in 2014 and 67 days in 2013. **Starbucks'** inventory turnover rates slowed considerably in 2012 and 2013, after it took over from Kraft all of the distribution of its products to grocery stores, warehouse clubs, and foodservice distributors. Since then, **Starbucks'** has significantly improved the efficiency of its inventory management. We project **Starbucks'** inventory turnover days will remain steady at 56 days. The projected year-end inventory amounts follow:

Inventory	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Costs of sales and occupancy costs	$7,787.5	$8,653.5	$9,390.4	$10,336.7	$11,358.7	$12,461.9
Costs of sales per day	$ 21.3	$ 23.7	$ 25.7	$ 28.3	$ 31.1	$ 34.1
Projected inventory turnover (in days)	56.0	56.0	56.0	56.0	56.0	56.0
Projected inventory balances	$1,306.4	$1,327.7	$1,440.7	$ 1,585.9	$ 1,742.7	$ 1,912.0

For some firms, such as retail chains, inventory is a large proportion of total assets. For such firms, you should link inventory forecasts to projections of the number of stores that will be operating in future years (or even, more specifically, to the number of square feet of retail space). For retail firms that operate large big-box stores (**Walmart** or **Costco**, for example), inventory projections may grow stepwise because each new store will require millions of dollars of additional inventory. Retail chains with seasonal sales will strive to have new stores (and thus new inventory) in place before heavy selling seasons (such as the back-to-school season for casual clothing and the Christmas season for toys); thus, analysts link inventory forecasts to projections of new stores in advance of these heavy selling seasons.

Projecting Prepaid Expenses and Other Current Assets

Prepaid expenses and other current assets represent items such as prepaid rent, advertising, and insurance. These items often vary in relation to the level of operating activity, such as sales, advertising, production, new stores or restaurants, and total assets. For **Starbucks**, many of these expenses vary with new company-operated stores. We will therefore project prepaid expenses and other current assets will grow in the future at the rate of growth in company-operated stores. The projected amounts are as follows:

Prepaid Expenses and Other Current Assets	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Company-operated stores	12,235	12,885	13,535	14,185	14,835	15,485
Growth rate		5.3%	5.0%	4.8%	4.6%	4.4%
Prepaid expenses and other current assets	$ 334.2	$ 352.0	$ 369.7	$ 387.5	$ 405.2	$ 423.0

Projecting Current and Noncurrent Deferred Income Tax Assets

Current and noncurrent deferred tax assets (and liabilities) are driven by temporary differences between accrued income tax expenses and income taxes paid to the governmental taxing authorities. These are complex, as discussed earlier in Chapter 6. For our purposes here, we will avoid getting too deep into the details of these accounts. However, for firms for which deferred tax assets and liabilities are very important, forecasts of these accounts may warrant more careful treatment.

At the end of fiscal 2015, **Starbucks** reports (net) deferred tax assets, with $381.7 million as current assets and $828.9 million as noncurrent assets. **Starbucks'** net deferred tax assets jumped considerably in 2013, when it recognized over $1.0 billion in new deferred tax assets arising from the litigation settlement with Kraft. In the Annual Report, Note 13, "Income Taxes," **Starbucks** does not disclose any information about when these deferred tax assets will be realized in future tax savings. It is therefore very difficult to predict changes in **Starbucks'** tax-paying status and realizations of tax savings from these deferred tax asset accounts. For simplicity, we will make the (somewhat arbitrary) assumptions that the current deferred tax assets will be fully realized in Year +1 and that the noncurrent deferred tax assets will decline by 10.0% per year during the forecast horizon. Going forward, we would monitor these accounts carefully, and be prepared to adjust these assumptions if **Starbucks'** tax status changes. The projected amounts are:

Current and Noncurrent Deferred Tax Assets	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Current	$381.7	$ 0.0	$ 0.0	$ 0.0	$ 0.0	$ 0.0
		Assume current tax benefits fully realized in Year +1.				
Noncurrent	$828.9	$746.0	$671.4	$604.3	$543.8	$489.5
Growth rate		(10.0%)	(10.0%)	(10.0%)	(10.0%)	(10.0%)

Projecting Long-Term Investments

In 2014 and 2015, **Starbucks'** long-term investments have been fairly steady, at $318.4 million and $312.5 million, respectively. Going forward, we will simply assume that **Starbucks'** long-term investments balances will grow by 3% per year. Later in this chapter we will include on the forecasted income statements the future interest income that we expect the cash, short-term, and long-term investments to generate. The projected balances follow:

Long-Term Investments	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Projected balances	$312.5	$321.9	$331.5	$341.5	$351.7	$362.3
Projected growth rate		3.0%	3.0%	3.0%	3.0%	3.0%

Projecting Equity and Cost Investments

Recall that we developed our forecast projections for equity investments when we projected expected income from equity investees on the income statement. Those projections are on page 661.

Projecting Property, Plant, and Equipment and Accumulated Depreciation

Recall that we developed our forecast projections for PP&E and accumulated depreciation when we projected depreciation expense on the income statement. Those projections are on page 659.

Projecting Other Long-Term Assets, Other Intangible Assets, and Goodwill

At the end of fiscal 2015, the majority of **Starbucks'** intangible assets involved goodwill ($1,575.4 million). Under U.S. GAAP and IFRS, a firm can only recognize goodwill as an asset when it acquires another company. Goodwill represents the portion of the acquisition price that the firm cannot allocate to other tangible or intangible assets. As **Starbucks** discloses in Note 8, "Other Intangible Assets and Goodwill," goodwill jumped by over $800 million in 2015 as a consequence of **Starbucks** acquiring and consolidating **Starbucks** Japan. U.S. GAAP and IFRS do not require firms to amortize goodwill because it has an indefinite useful life, but firms must test goodwill values annually for impairment and write the carrying values down to fair value if deemed impaired. Thus far, **Starbucks** has deemed it necessary to recognize only very minor impairment losses on its goodwill (a total of only $9.9 million in accumulated impairment charges).

On the 2015 balance sheet, **Starbucks** recognizes other intangible assets, including $302.5 million in finite-lived intangibles (primarily acquired rights) and $217.9 million in indefinite-lived intangibles (primarily trade names, trademarks, and patents). **Starbucks** also recognizes $415.9 million in miscellaneous other long-term assets.

Because of their nature, these various types of assets, especially goodwill and intangibles, tend to grow with company acquisitions, as was the case in 2015. Absent announcements or disclosures about pending acquisitions, these types of transactions are inherently hard to forecast accurately. Therefore, for simplicity, we project that goodwill, other intangible assets, and other long-term assets will grow by 5% per year. We will also assume that no future impairment charges will be necessary for these assets. The projected amounts are as follows (allow for rounding):

Other Long-Term Assets, Other Intangible Assets, and Goodwill	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Other long-term assets	$ 415.9	$ 436.7	$ 458.5	$ 481.5	$ 505.5	$ 530.8
Expected growth rates		5.0%	5.0%	5.0%	5.0%	5.0%
Other intangible assets	$ 520.4	$ 546.4	$ 573.7	$ 602.4	$ 632.5	$ 664.2
Expected growth rates		5.0%	5.0%	5.0%	5.0%	5.0%
Goodwill	$1,575.4	$1,654.2	$1,736.9	$1,823.7	$1,914.9	$2,010.7
Expected growth rates		5.0%	5.0%	5.0%	5.0%	5.0%

Projecting Assets as a Percentage of Total Assets

In some circumstances, you may need to project individual asset amounts that will vary as a percentage of total assets, particularly for firms that maintain a steady proportion of total assets invested in specific types of assets. For example, suppose a firm's strategy is to maintain 7.0% of total assets in cash for liquidity purposes. Also suppose our projected amounts for Year +1 for all of the individual assets *other than cash* amount to $100.0 million. The $100 million subtotal represents 93.0% (1.00 − 0.07) of total assets. Therefore, projected total assets should equal $107.527 million ($100.0 million/0.93).

Thus, in this hypothetical example, the ending cash balance would equal \$7.527 million ($0.07 \times$ \$107.527 million).

Note that this approach to forecasting introduces some circularity into the projected financial statements; the cash balance is a function of total assets, which is a function of the cash balance. This forecast approach is not unrealistic, nor does it create a problem for the computations. A later subsection of this chapter discusses how to solve for such codetermined variables in financial statement forecasts.

Projected Total Assets

We have now projected amounts for each of the assets reported on **Starbucks'** balance sheets. Our projected asset totals are as follows:

	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Projected total assets	\$12,446.1	\$12,829.0	\$13,540.1	\$14,495.5	\$15,597.5	\$17,552.1

Next, we will turn to forecasting operating liabilities, as well as debt and equity financing.

Projecting Accounts Payable

Starbucks reports \$684.2 million in accounts payable on its 2015 balance sheet. Future credit purchases of inventory and **Starbucks'** payment policy to its suppliers will drive future accounts payable. The ratios discussed in Chapter 4 and reported in the Analysis spreadsheet of FSAP show that **Starbucks** maintains a very stable payables policy. Days payable have varied within a narrow range from 26 to 28 days over the past five years. We assume that **Starbucks** will continue to maintain an accounts payable period of 28 days in the future. To forecast future accounts payable balances, we begin by forecasting inventory purchases, which drive payables. We rely on our prior forecasts of **Starbucks'** cost of sales (net of the occupancy costs) and add the changes in the inventory balances to compute inventory purchases, which will flow through accounts payable. We then project ending balances in accounts payable, assuming 28-day payable periods.

As discussed earlier, in Year +1 the projected cost of sales and occupancy costs is \$8,653.5 million (page 656).[15] This is the cost of the inventory **Starbucks** will have to purchase simply to replenish the inventory we project it will sell in Year +1. However, in our inventory forecasts (page 672), we projected that **Starbucks'** inventory balance will grow to \$1,327.7 million by the end of Year +1, an increase of \$21.3 million. This represents additional inventory purchases. The total projected inventory purchases in Year +1 will be \$8,674.8 million (\$8,653.5 million + \$21.3 million), all of which will be on credit terms, driving accounts payable. On an average day during Year +1, **Starbucks** will purchase \$23.8 million in inventory, and incur \$23.8 million in accounts payable. If **Starbucks** maintains a 28-day payables policy, then the accounts payable balance will be $665.5 million ($23.8 million × 28.0 days). The accounts payable projections for Year +1 to +5 are as follows (allow for rounding):

Accounts Payable	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Costs of sales and occupancy costs	$7,787.5	$8,653.5	$9,390.4	$10,336.7	$11,358.7	$12,461.9
Projected inventory balances	$1,306.4	$1,327.7	$1,440.7	$ 1,585.9	$ 1,742.7	$ 1,912.0
Projected changes in inventory		$ 21.3	$ 113.1	$ 145.2	$ 156.8	$ 169.3
Projected inventory purchases		$8,674.8	$9,503.4	$10,481.9	$11,515.5	$12,631.2
Average inventory purchases per day		$ 23.8	$ 26.0	$ 28.7	$ 31.5	$ 34.6
Projected turnover (in days)	28.0	28.0	28.0	28.0	28.0	28.0
Accounts payable	$ 684.2	$ 665.5	$ 729.0	$ 804.1	$ 883.4	$ 969.0

Projecting Accrued Liabilities

In **Starbucks'** 2015 Annual Report, Note 8, "Supplemental Balance Sheet Information" (Appendix A), discloses that at the end of 2015 accrued liabilities consisted of amounts payable for various activities, including compensation and benefits, occupancy costs, taxes, dividends, and other expenses. In recent years, accrued liabilities have been growing at roughly the same rate of growth as revenues (and, in some years, faster). In 2015, for example, revenues grew by 16.5% and accrued liabilities grew by 16.3%. We therefore forecast that accrued liabilities will grow with revenues (allow for rounding).

Accrued Liabilities	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Accrued liabilities	$1,760.7	$1,991.8	$2,174.9	$2,416.0	$2,679.2	$2,966.6
Growth with revenue growth rates		13.1%	9.2%	11.1%	10.9%	10.7%

Projecting Insurance Reserves

In **Starbucks'** 2015 Annual Report, Note 1, "Summary of Significant Accounting Policies" (Appendix A), explains that the firm uses a combination of insurance and self-insurance mechanisms to provide for potential liabilities for certain risks, such as workers' compensation, heath care benefits, general liability, and others. We will simply forecast that insurance reserves will grow 3% per year (allow for rounding).

Insurance Reserves	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Insurance reserves	$224.8	$231.5	$238.5	$245.6	$253.0	$260.6
Growth rates		3.0%	3.0%	3.0%	3.0%	3.0%

Projecting Stored-Value Card Liabilities

Starbucks' stored-value cards can be loaded at company-operated stores, most licensed stores, and online. When a customer loads a card, the customer pays **Starbucks** cash, and the stored-value card can then be redeemed at **Starbucks'** locations for products and services. The stored-value card liability is therefore a service obligation to customers for prepaid revenues. At the end of fiscal 2015, the stored-value card liability amounted to $983.8 million. We forecast that stored-value card liabilities will grow with revenues (allow for rounding).

Stored-Value Card Liabilities	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Stored-value card liabilities	$983.8	$1,113.0	$1,215.2	$1,349.9	$1,497.0	$1,657.6
Growth with revenues		13.1%	9.2%	11.1%	10.9%	10.7%

Projecting Other Long-Term Liabilities

Other long-term liabilities are accrued liabilities for expenses that relate to pension obligations, health care obligations, long-term compensation, and other operating and administrative expenses. We therefore project other noncurrent liabilities will grow with general and administrative expenses, which we assume grow with revenues (allow for rounding):

Other Long-Term Liabilities	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Other long-term liabilities	$625.3	$707.4	$772.4	$858.0	$951.5	$1,053.6
Growth with revenues		13.1%	9.2%	11.1%	10.9%	10.7%

Step 4: Project Financial Leverage, Financial Assets, Common Equity Capital, and Financial Income and Expense Items

LO 10-2d

Build forecasts of future balance sheets, income statements, and statements of cash flows by applying the seven-step forecasting framework to project: d. financial leverage, capital structure, and financial income items.

After completing forecasts of the operating assets and liabilities of the balance sheet, we must now project any financial assets the firm will hold, and the financial debt and shareholders' equity amounts that will be necessary to finance the firm's operating and investing activities. In addition, we project the effects of investing and financing on net income by projecting future interest income, interest expense, and other elements of financial income.

For firms that regularly maintain a particular capital structure over time, you can use the common-size balance sheet percentages to project amounts of debt and equity capital. If the firm has a target capital structure that consists of stable proportions of liabilities and equity (for instance, 60% liabilities and 40% equity), you can use these common-size percentages and the projected amounts of total assets to project future totals of liabilities and equity.

Alternatively, you can project debt capital and shareholders' equity accounts by projecting potential future changes in the financial leverage strategy of the firm. For instance, in recent years, interest rates on corporate debt have been at historically low levels, so many firms have been taking on more financial leverage with greater amounts of short- and long-term debt and using this debt financing to reduce shareholders' equity through repurchases of common shares and increased dividends. In other cases, you may need to project how a firm will alter its capital structure as a result of a merger, acquisition, or divestiture transaction.

In this section, we forecast debt and equity by projecting the financial leverage strategy of **Starbucks** over our forecast horizon. We discuss each account in turn.

Projecting Financial Assets

You must assess the firm's business activities and financial strategy to determine the extent to which the firm uses financial assets for operating liquidity purposes versus financial purposes. Most firms use financial assets, such as cash and short-term and long-term investment securities, to accomplish one or all of the following:

- manage seasonal swings in operating liquidity.
- provide a financial cushion for future uncertainties.
- have financial flexibility to take advantage of profitable opportunities when they arise.

As such, for these firms it makes sense to forecast financial assets as part of the liquidity and operating activities of the firm. For example, **Starbucks'** 2015 balance sheet recognizes cash, short-term investments, long-term investments, and equity investments. As discussed previously, **Starbucks** uses cash and short-term and long-term investments to provide liquidity for operating activities. The equity investments represent investments in joint ventures closely related to **Starbucks'** operating activities. Therefore, we included these types of financial assets in our projections of **Starbucks'** operating activities.

By contrast, some firms hold excess cash and/or investment securities that are not needed for operating liquidity, and are instead intended for future financial purposes, such as debt retirement, corporate acquisitions, repurchase of shares, or payment of dividends. When forecasting the future financial capital structure for firms like these, you must project the future financial assets that represent financial savings that can be used for debt retirement or other financial purposes. Suppose, for example, a firm had issued bonds to finance plant and equipment and the bond indenture agreement required the firm to maintain a bond sinking fund (a reserve of cash and securities to be used for future bond retirement). The cash and securities in the sinking fund would represent financial assets intended for debt retirement and should be projected with the firm's financial structure rather than as part of the firm's operating activities. As of the 2015 balance sheet, **Starbucks** does not report any short-term or long-term investment securities for debt retirement purposes, and because of its ability to generate cash flows from operations, **Starbucks** is not likely to need future reserves for debt retirement.

Projecting Short-Term and Long-Term Debt

As of the end of 2015, **Starbucks** had issued a total of $2,347.5 million in long-term debt. The common-size balance sheet for **Starbucks** (Chapter 1) shows that long-term debt has become an increasingly larger proportion of total assets. From 2013 to 2015, the proportion of long-term debt in **Starbucks'** capital structure jumped considerably, from roughly 11% of total assets to 19%.

Starbucks' statement of cash flows for 2015 (Appendix A) indicates the firm generated $3,749.1 million of net cash flow from operating activities and used $1,520.3 million for investing activities. In terms of financing activities in 2015, **Starbucks** used $928.6 million in cash to pay dividends and $1,436.1 million to repurchase common stock, after issuing $848.5 million in long-term debt and paying down roughly $610.1 million in long-term debt. Clearly, **Starbucks** generates a lot of cash from its operating activities and is choosing to increase its debt obligations in order to return more cash to shareholders through dividends and share repurchases. **Starbucks** appears to be shifting its financial leverage strategy to recapitalize by issuing greater amounts of long-term debt while paying larger dividends and repurchasing more common equity shares.

In fact, during the first half of fiscal 2016 (while this chapter was being written), **Starbucks** issued a total of nearly $1.0 billion of new long-term debt obligations. We include this new additional level of long-term debt in our forecasts for Year +1.

Starbucks' outstanding long-term debt matures at varying dates extending from 2016 to 2045. Note 9, "Debt" (Appendix A), discloses information on the maturities and reveals that $400 million of long-term debt will mature in December 2016 (fiscal

2017; Year +2), and another $350 million will mature in December 2018 (fiscal 2019; Year +4). In our forecasts, we therefore reduce long-term debt by these amounts the year before they mature, and include them in the current liability account "Current Maturities of Long-Term Debt." In the years in which they mature, we forecast that they will be repaid.[16] To keep long-term debt levels stable, we will also project that in the years in which long-term debts mature, **Starbucks** will replace them with new long-term debt issues.

The projected amounts for current maturities and long-term debt are as follows:

Long-Term Debt and Current Maturities of Long-Term Debt	Year +1	Year +2	Year +3	Year +4	Year +5
Long-term debt					
Beginning balance	$2,347.5	$2,947.5	$3,347.5	$2,997.5	$3,347.5
Add: New issues	$1,000.0	$ 400.00	$ 0.0	$ 350.0	$ 0.0
Less: Maturities	$ (400.0)	$ 0.0	$ (350.0)	$ 0.0	$ 0.0
Ending balance	$2,947.5	$3,347.5	$2,997.5	$3,347.5	$3,347.5
Current maturities of long-term debt					
Beginning balance	$ 0.0	$ 400.0	$ 0.0	$ 350.0	$ 0.0
Add: Maturing issues	$ 400.0	$ 0.0	$ 350.0	$ 0.0	$ 0.0
Less: Repayments	$ 0.0	$ (400.0)	$ 0.0	$ (350.0)	$ 0.0
Ending balance	$ 400.0	$ 0.0	$ 350.0	$ 0.0	$ 0.0

We will use these projected amounts for interest-bearing debt to project **Starbucks'** future interest expense.

Projected Total Liabilities

We have now made projections of all of **Starbucks'** liabilities. Combining them, our projected liabilities for **Starbucks** over the forecast horizon are as follows:

Total Liabilities	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Accounts payable	$ 684.2	$ 665.5	$ 729.0	$ 804.1	$ 883.4	$ 969.0
Accrued liabilities	1,760.7	1,991.8	2,174.9	2,416.0	2,679.2	2,966.6
Insurance reserves	224.8	231.5	238.5	245.6	253.0	260.6
Stored-value card liabilities	983.8	1,113.0	1,215.2	1,349.9	1,497.0	1,657.6
Current maturities of long-term debt	0.0	400.0	0.0	350.0	0.0	0.0
Current liabilities	$3,653.5	$4,401.8	$4,357.6	$5,165.6	$5,312.6	$ 5,853.7
Long-term debt	2,347.5	2,947.5	3,347.5	2,997.5	3,347.5	3,347.5
Other long-term liabilities	625.3	707.4	772.4	858.0	951.5	1,053.6
Total liabilities	$6,626.3	$8,056.7	$8,477.5	$9,021.1	$9,611.6	$10,254.8
Total liabilities as a percent of total assets	53.2%	62.8%	62.9%	63.7%	65.0%	63.7%

Projecting Interest Expense

We can now project **Starbucks'** interest expense, based on our projected balances in interest-bearing debt. Note 9, "Debt" (Appendix A), discloses the amounts, maturities, and interest rates on the debt issues outstanding. It indicates that at the end of 2015, the interest rates ranged from 0.875% on notes maturing in 2016 to 4.3% on notes maturing in 2045. In addition, the two most recent debt issues are $500 million in 2.1% notes maturing in 2021 and $500 million in 2.45% notes maturing in 2026. We can use all of these disclosures to estimate the weighted-average interest rate on total debt outstanding, as follows:

Long-Term Debt and Weighted-Average Interest Rates

Debt Issues and Maturities	Face Value Amounts	Proportion of Total	Stated Interest Rates	Weighted Average Interest Rates
2016 Notes	$ 400.0	0.119	0.875%	0.1045%
2018 Notes	$ 350.0	0.104	2.000%	0.2090%
2021 Notes	$ 500.0	0.149	2.100%	0.3134%
2022 Notes	$ 500.0	0.149	2.700%	0.4030%
2023 Notes	$ 750.0	0.224	3.850%	0.8619%
2026 Notes	$ 500.0	0.149	2.450%	0.3657%
2045 Notes	$ 350.0	0.104	4.300%	0.4493%
Totals	**$3,350.0**	1.000		**2.7067%**

We therefore project interest expense using the weighted-average interest rate of 2.7067% on the average amount of interest-bearing debt in Year +1 through Year +5 to match the weighted-average interest rate on the outstanding debt. Using the projected amounts of total interest-bearing debt in the prior section, the projected interest expense amounts follow (allow for rounding):

Interest-Bearing Debt and Interest Expense	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Long-term debt	$2,347.5	$2,947.5	$3,347.5	$2,997.5	$3,347.5	$3,347.5
Current maturities of long-term debt	$ 0.0	$ 400.0	$ 0.0	$ 350.0	$ 0.0	$ 0.0
Total interest-bearing debt	$2,347.5	$3,347.5	$3,347.5	$3,347.5	$3,347.5	$3,347.5
Average interest-bearing debt		$2,847.5	$3,347.5	$3,347.5	$3,347.5	$3,347.5
Weighted average interest rate		2.7067%	2.7067%	2.7067%	2.7067%	2.7067%
Projected interest expense	$ 70.5	$ 77.1	$ 90.6	$ 90.6	$ 90.6	$ 90.6

The interest expense projections are higher than recent past interest expense amounts for **Starbucks**, reflecting **Starbucks'** shift in financial strategy in 2015 to greater reliance on long-term debt. We can now enter these interest expense amounts in the projected income statements. If the projected balance sheets imply that **Starbucks** will need larger or smaller amounts of long-term debt to finance future asset growth, then we will need to recompute the interest expense projections to reflect different amounts of debt.

Projecting Interest Income

We can also project our first-pass estimates of **Starbucks'** interest income on financial assets, such as cash and short-term and long-term investments. In 2015, **Starbucks** recognized \$43.0 million in interest and other income.[17] The average amount of interest-earning assets (cash and short-term and long-term investments) during 2015 was \$2,043.0 million [(\$1,530.1 + \$1,708.4 + \$81.3 + \$135.4 + \$312.5 + \$318.4)/2], for an average return of 2.1% (\$43.0/\$2,043.0). This low average rate of return reflects the very low interest rate environment present during 2015. In addition, it is likely that **Starbucks'** cash and investment securities are low-risk but liquid instruments, and therefore yield low rates of return. At the beginning of Year +1, yields on low-risk, highly liquid short-term and medium-term U.S. Treasury bonds were in the 1.0% to 2.0% range. Assuming interest rates and yields remain stable over the forecast horizon, we project **Starbucks** will earn a 2.1% return on the average balances in cash and short-term and long-term investments each year. The projected amounts for interest income are as follows (allow for rounding):

Interest-Earning Assets and Interest Income	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Cash	\$1,530.1	\$ 1,781.8	\$ 1,945.5	\$ 2,161.2	\$ 2,396.7	\$ 2,653.7
Short-term investments	\$ 81.3	\$ 83.7	\$ 86.3	\$ 88.8	\$ 91.5	\$ 94.2
Long-term investments	\$ 312.5	\$ 321.9	\$ 331.5	\$ 341.5	\$ 351.7	\$ 362.3
Total interest-earning assets	\$1,923.9	\$ 2,187.4	\$ 2,363.3	\$ 2,591.5	\$ 2,839.9	\$ 3,110.2
Average interest-earning assets		\$2,055.65	\$2,275.35	\$2,477.39	\$2,715.68	\$2,975.06
Weighted-average interest rate		2.1%	2.1%	2.1%	2.1%	2.1%
Projected interest income	\$ 43.0	\$ 43.2	\$ 47.8	\$ 52.0	\$ 57.0	\$ 62.5

If the projected balance sheets imply that **Starbucks** will generate and retain larger amounts of cash and investment securities, then we will recompute the interest income projections to reflect additional interest-earning assets.

Projecting Noncontrolling Interests

Noncontrolling interests in equity are similar to the equity method investments (investments in noncontrolled affiliates) described earlier, but in this case the firm is the controlling shareholder and consolidates the subsidiaries. Noncontrolling interests therefore account for the equity ownership of minority investors that own less than 50% of the shares of the subsidiary company (this is the mirror image of accounting for investments in equity affiliates).

At the end of 2015, **Starbucks** has only \$1.8 million of equity capital from noncontrolling interest shareholders. These investors own significant but not controlling proportions of the equity in certain subsidiaries that **Starbucks** controls and consolidates.

We should project the future amounts of noncontrolling equity by projecting additional investments by or retirements of these noncontrolling equity investors as well as the proportionate share of income and dividends from these subsidiaries to these investors. In addition, we should reduce net income in our projected income statements by the amount of net income attributable to noncontrolling interests each year, with the remainder of net income attributable to **Starbucks'** shareholders.

Because the amount of noncontrolling interests in equity are so minor (0.03% of total shareholders' equity), for simplicity we will assume that **Starbucks** will retire all of the noncontrolling interests in Year +1, and that **Starbucks'** subsidiaries will not raise any additional noncontrolling equity capital during the forecast horizon.

Projecting Common Stock, Preferred Stock, and Additional Paid-in Capital

As Chapter 7 explains, common stock, preferred stock, and paid-in capital accounts increase as the firm raises capital by selling common and/or preferred shares to investors or by issuing shares in a merger or acquisition. These accounts decrease when the firm retires shares. The paid-in capital accounts also increase when firms award share-based compensation and bonuses, such as stock options or restricted share units, to employees, executives, and board members. Paid-in capital accounts increase (but retained earnings decrease) by the fair value of the share-based compensation expense. Subsequently, if individuals exercise stock options or convert restricted shares into unrestricted shares, then paid-in capital accounts increase by the net amount of any cash received from the individuals (such as the exercise price of the options).

Starbucks' consolidated statement of equity (Appendix A) shows that, in 2015, **Starbucks'** has no preferred stock outstanding. We will assume that **Starbucks** will not issue any preferred shares during the forecast horizon.

The statement of equity also shows that **Starbucks** has 1,485.1 million common shares outstanding with a par value of $0.001 per share, so the common stock at par account is only $1.5 million at the end of fiscal 2015. [Note: **Starbucks** had a 2-for-1 stock split during fiscal 2015, doubling the number of outstanding shares.] The additional paid-in capital account amounts to only $41.1 million at the end of fiscal 2015. The common-sized balance sheet data (Chapter 1) show that **Starbucks'** paid-in capital accounts (par value plus additional paid-in capital) in 2015 amount to only 0.342% of total assets. Given that **Starbucks** has shifted its capital strategy to increase the proportion of long-term debt, we do not expect significant future issues of common equity. We therefore simply project common stock and additional paid-in capital will grow with total assets, remaining at 0.342% of total assets. The projected amounts for common stock and additional paid-in capital are as follows:

Common Stock and Additional Paid-in Capital	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Total assets	$12,446.1	$12,829.0	$13,540.1	$14,495.5	$15,597.5	$17,552.1
Paid-in capital as a percent of total assets	0.342%	0.342%	0.342%	0.342%	0.342%	0.342%
Projected common stock and additional paid-in capital	$ 42.6	$ 43.9	$ 46.3	$ 49.6	$ 53.4	$ 60.1

Projecting Accumulated Other Comprehensive Income or Loss

As described in Chapters 8 and 9, under U.S. GAAP, other comprehensive income items result from four specific types of asset and liability revaluations: fair value gains and losses on available-for-sale securities, foreign currency translation adjustments, adjustments to certain pension and retiree benefit obligations, and fair value gains and losses on cash flow hedges. Because economywide changes in interest rates and foreign exchange rates tend to be transitory and because many firms tend to hedge or mitigate exposure to such risks, it is often very difficult to predict with confidence whether a particular firm will generate persistent gains or losses from such changes over long periods of time. As such, analysts commonly forecast gains or losses from other comprehensive income items to be zero, on average.

On the consolidated statement of comprehensive income (Appendix A), **Starbucks** reported net income of $2,759.3 million for 2015 but comprehensive income of $2,534.6 million—**Starbucks'** other comprehensive income items amounted to a loss of $224.7 million after tax. As that statement shows, these income items were attributable primarily to foreign currency translation adjustments. In 2015, the U.S. dollar's value strengthened against many other currencies in which **Starbucks** holds net assets, which triggered the –$222.7 million translation adjustment. Also in 2015, **Starbucks** recognized various small amounts for fair value gains and losses on available-for-sale securities, and cash flow and net investment hedges. In 2015, **Starbucks** also reclassified net (after-tax) losses of $43.6 million in earnings. According to **Starbucks'** consolidated statement of equity (Appendix A), accumulated other comprehensive income/loss was $25.3 million at the beginning of 2015, and, consequently, by the end of 2015 it had become −$199.4 million.

In prior years, **Starbucks'** other comprehensive income items were much smaller. **Starbucks** recognized other comprehensive losses of only –$41.7 million in fiscal 2014, and other comprehensive income of $44.3 million in fiscal 2013. For simplicity, we project that **Starbucks** will experience neither persistent income nor losses from other comprehensive income items in the future. We are essentially assuming that **Starbucks'** future foreign currency translation adjustments are equally likely to be positive or negative in any given year and that, on average, they will be zero over time. In addition, we assume that available-for-sale investment securities and cash flow hedges are equally likely to generate fair value gains or losses in any given year and that, on average, they will be zero over the forecast horizon. Therefore, we project that accumulated other comprehensive loss will remain at its current level. Accordingly, other comprehensive income items will also be zero in future years.

Step 5: Project Provisions for Taxes, Net Income, Dividends, Share Repurchases, and Retained Earnings

Thus far we have developed forecasts of **Starbucks'** operating activities, including operating income as well as the operating assets and liabilities. We have projected **Starbucks'** future financial liabilities, common equity capital, and financial income items including interest income, interest expense, and income from noncontrolled affiliates. In Step 5, we complete the forecasting of **Starbucks'** net income, dividends, share repurchases, and retained earnings. This will lead us into Step 6, in which we make our balance sheet forecasts balance.

Projecting Provisions for Income Taxes

As Chapter 9 discusses, **Starbucks'** Note 13, "Income Taxes" (Appendix A), shows the reconciliation between the statutory tax rate and **Starbucks'** average, or effective, tax rate. The statutory U.S. federal income tax rate was 35.0% during 2013–2015. During 2015, **Starbucks** experienced an effective income tax rate of only 29.3%. This relatively low effective tax rate in 2015 includes an additional 2.8% tax that **Starbucks** paid for state income taxes. However, **Starbucks'** effective income tax rate was offset by various tax benefits, including a decrease of 2.1% from lower tax rates in international tax jurisdictions, a benefit of 2.2% from the domestic production activity deduction, a benefit of 3.7% from the tax-deferred gain on the acquisition of the joint venture in Japan, and various other factors. In 2014, **Starbucks** experienced a more normal effective tax rate of 34.6%.

As we noted earlier, **Starbucks'** 2015 Annual Report, the MD&A section titled "Fiscal 2016—The View Ahead," and subsequent quarterly disclosures early in 2016 provide information from **Starbucks'** managers about their expectations. In these disclosures, **Starbucks** reveals it expects an effective tax rate of roughly 34.0% in Year +1. We therefore assume that **Starbucks'** average combined federal, state, and foreign income tax rate for Year +1 through Year +5 will be 34.0%. The projected amounts for income tax expense are reported in the table in the next section.

Also discussed earlier, Note 13 discloses that **Starbucks** has net deferred tax assets of $1,137.4 million at the end of fiscal 2015, a large portion of which stems from deferred tax savings from the litigation settlement with Kraft. When these deferred tax assets are realized, they will enable **Starbucks** to generate future tax savings. Although these deferred tax assets will not reduce **Starbucks'** expected future effective tax rate, they will allow **Starbucks** to pay lower amounts of cash taxes. As we reduce our projections of these deferred tax assets on our projected balance sheets, we will also capture the cash tax savings on our projected statements of cash flows over our forecast horizon.

Net Income Attributable to Starbucks' Common Shareholders

All of the elements of the income statement, including first-iteration estimates of interest expense, interest income, and income taxes, are now complete. Recall that Exhibit 10.3 presents these income statement projections. The projected amounts of net earnings, net profit margins, and the implied earnings growth rates are as follows:

Net Income Projections	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Projected total net revenues	$19,162.7	$21,678.3	$23,670.5	$26,294.2	$29,159.4	$32,286.7
Projected operating income	$ 3,601.0	$ 4,267.7	$ 4,710.4	$ 5,326.4	$ 6,002.9	$ 7,444.8
Projected interest income	43.0	43.2	47.8	52.0	57.0	62.5
Projected interest expense	(70.5)	(77.1)	(90.6)	(90.6)	(90.6)	(90.6)
Gain from acquisition of joint venture	390.6	0.0	0.0	0.0	0.0	0.0
Loss on extinguishment of debt	(61.1)	0.0	0.0	0.0	0.0	0.0
Income before taxes	$ 3,903.0	$ 4,233.8	$ 4,667.5	$ 5,287.8	$ 5,969.3	$ 7,416.6
Effective tax rates	(29.3%)	(34.0%)	(34.0%)	(34.0%)	(34.0%)	(34.0%)
Income tax expense	(1,143.7)	(1,439.5)	(1,587.0)	(1,797.9)	(2,029.6)	(2,521.7)
Projected net earnings	$ 2,759.3	$ 2,794.3	$ 3,080.6	$ 3,490.0	$ 3,939.8	$ 4,895.0
Net profit margin	14.4%	12.9%	13.0%	13.3%	13.5%	15.2%
Net income growth rates	33.4%	1.3%	10.2%	13.3%	12.9%	24.2%

The forecasts of net earnings for **Starbucks** imply a slightly lower profit margin than in 2013–2015 and only modest growth in net income in Year +1, in part because of the increase in the expected tax rate. Profit margins and earnings growth rates return to higher, more normal levels for **Starbucks** in Years +2 through +5.

Projecting Dividends and Share Repurchases

Profitable, mature companies like **Starbucks** can distribute excess cash to shareholders through dividend payments as well as by repurchasing common shares. In recent years, given its profitability and strong cash flows, **Starbucks** has been increasing these cash flows to its shareholders. Dividend payout rates have increased from $0.445 per share in 2013, to $0.550 per share in fiscal 2014, to $0.680 per share in fiscal 2015. For Year +1, **Starbucks'** board of directors approved a further increase to dividend payments of $0.800 per share. Relying on that announcement, and with roughly 1,485 million shares outstanding, it suggests **Starbucks** will pay roughly $1,188.0 million in dividends in Year +1.[18] Given that we forecast **Starbucks** will generate $2,794.3 million in net earnings in Year +1, it implies **Starbucks** will pay out 42.5% of earnings in dividends. We will project that **Starbucks'** dividend payout policy will average 42.5% of projected net earnings in Years +1 through + 5. Therefore, forecasts of dividends to common shareholders will vary over time with net earnings.

In addition to paying dividends, **Starbucks** also distributes large amounts of cash to shareholders through repurchases of common shares. For some firms, repurchases of common shares are accounted for in a treasury stock account, which is a negative (contra) account in shareholders' equity on the balance sheet. The treasury stock account becomes *more* negative when the firm repurchases some of its outstanding common equity shares. The treasury stock account becomes *less* negative when the firm reissues treasury shares on the open market, uses them to meet stock option exercises, issues them in merger or acquisition transactions, or retires them. (See Chapter 7 for more discussion of accounting for treasury stock transactions.)

Starbucks, however, is incorporated in the state of Washington, and state laws require that it subtract amounts paid for common share repurchases from retained earnings.[19] In 2013 through 2015, **Starbucks'** statement of common shareholders' equity (Appendix A) reveals that it has been using increasing amounts of cash to repurchase common shares: $544.1 million in 2013, $769.8 million in 2014, and $1,431.8 million in 2015.[20]

In **Starbucks'** 2015 Annual Report, the MD&A section titled "Financial Condition, Liquidity, and Capital Resources" (Appendix B, which can be found online at the book's companion website at www.cengagebrain.com) discloses that in July 2015, **Starbucks** announced the board of directors had approved an increase of 50 million shares to the ongoing share repurchase program. **Starbucks** will likely continue to make substantial repurchases of common shares. We project that **Starbucks** will use $1,500 million in cash each year in Years +1 through +5 to repurchase common shares.

We may need to reduce the projected amount of share repurchases in particular years if we determine later in our analysis that **Starbucks** will not have sufficient cash flow for these repurchases, or if our equity valuation estimates indicate that the capital market has overpriced **Starbucks'** shares. Alternately, we may increase these projected amounts if our analysis reveals that **Starbucks** will have excess future cash flow or if our valuation estimates indicate **Starbucks'** shares are underpriced.

In projecting stock repurchases net of stock reissues, we assume that employees will continue to exercise stock options and other stock-based compensation awards in future years. We may need to revise this assumption later in the analysis if our equity valuation estimates indicate that **Starbucks'** stock options are not likely to be "in the money." We implicitly include an expense for the fair value of stock-based compensation in the projections of general and administrative expense.

The projected amounts for dividends and common share repurchases are as follows:

Dividends and Common Share Repurchases	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Projected net earnings	$2,759.3	$2,794.3	$3,080.6	$3,490.0	$3,939.8	$4,895.0
Dividend payout policy	36.8%	42.5%	42.5%	42.5%	42.5%	42.5%
Projected dividend payments	$1,016.2	$1,187.6	$1,309.2	$1,483.2	$1,674.4	$2,080.4
Repurchases of common shares	$1,431.8	$1,500.0	$1,500.0	$1,500.0	$1,500.0	$1,500.0

Retained Earnings

In general, retained earnings typically increase by the amount of net income and decrease for dividends declared and net losses. For **Starbucks**, amounts paid to repurchase common shares are subtracted from retained earnings, too. The implied changes in retained earnings are as follows (allow for rounding):

Retained Earnings	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Beginning balance	$ 5,974.8	$ 6,081.5	$ 6,352.9	$ 6,859.6	$ 7,625.0
Projected net earnings	2,794.3	3,080.6	3,490.0	3,939.8	4,895.0
Projected dividend payments	(1,187.6)	(1,309.2)	(1,483.2)	(1,674.4)	(2,080.4)
Projected common share Repurchases	(1,500.0)	(1,500.0)	(1,500.0)	(1,500.0)	(1,500.0)
Ending balance	$ 6,081.5	$ 6,352.9	$ 6,859.6	$ 7,625.0	$ 8,939.6

Step 6: Balance the Balance Sheet

LO 10-2f

Build forecasts of future balance sheets, income statements, and statements of cash flows by applying the seven-step forecasting framework to project: f. a balance sheet that balances.

Even though the first-pass forecasts of all amounts on the income statement and balance sheet are complete, the balance sheet does not balance because we have projected individual asset and liability accounts to capture their underlying business activities, which are not perfectly correlated. The difference between the initial projected total assets and the projected total liabilities and shareholders' equity each year represents the amount by which we must adjust a "flexible" financial account to balance the balance sheet. If the difference is a positive amount, projected assets exceed projected liability and equity claims, so the firm must raise additional debt or equity capital or reduce projected assets. If the difference is a negative amount, projected assets are less than projected liability and equity financing, in which case the firm can pay down debt, issue larger dividends, repurchase more shares, or increase investments in financial assets. The change in the difference represents the increment by which we must adjust the flexible financial account each year.

The analyst must evaluate the firm's financial flexibility and adjust the balance sheet accordingly. For some firms (for example, start-ups), financial flexibility may be in cash or marketable securities, which represent financial liquidity "safety valves." These firms often keep relatively large amounts of cash or marketable securities on the balance sheet for financial slack and use the funds when necessary to meet periodic cash requirements. For these firms, large inflows of cash (such as from a new issue of debt or equity shares) build up the cash and marketable securities accounts, and large outflows (such as for the purchase of an asset or R&D expenditures) deplete the accounts. For these firms, analysts can use cash or marketable securities as the financial flexibility account needed to balance the balance sheet after all other balance sheet amounts have been determined.

For profitable growth firms that do not have large reserves of excess cash or marketable securities, financial flexibility may be exercised through short-term or long-term debt or equity. As the firm grows and invests in increasing productive capacity, it must raise the necessary capital through borrowing or issuing equity. As the firm matures and becomes a cash cow, it will shift how it uses its financial flexibility to pay down debt and perhaps initiate or increase dividends and share repurchases. You should consider carefully what financial flexibility the firm has and is likely to use.

Balancing Starbucks' Balance Sheets

Currently, our projections of **Starbucks'** total assets minus total liabilities and shareholders' equity indicate the amounts by which our balance sheets do not balance. These amounts are shown below (allow for rounding):

Projected Balance Sheet Amounts	Year +1	Year +2	Year +3	Year +4	Year +5
Total assets (A)	\$12,829.0	\$13,540.1	\$14,495.5	\$15,597.5	\$17,552.1
Total liabilities	\$ 8,056.7	\$ 8,477.5	\$ 9,021.1	\$ 9,611.6	\$10,254.8
Common stock and additional paid-in capital	43.9	46.3	49.6	53.4	60.1
Accumulated other comprehensive loss	(199.4)	(199.4)	(199.4)	(199.4)	(199.4)
Retained earnings	6,081.5	6,352.9	6,859.6	7,625.0	8,939.6
Total liabilities and equity (L + E)	\$13,982.7	\$14,677.3	\$15,730.9	\$17,090.6	\$19,055.0
Difference: A − (L + E)	\$(1,153.7)	\$(1,137.2)	\$(1,235.4)	\$(1,493.0)	\$(1,503.0)
Change in the difference	\$(1,153.7)	\$ 16.5	\$ (98.2)	\$ (257.6)	\$ (9.9)

In Year +1, the first-iteration forecasts project that liabilities and equities exceed assets by $1,153.7 million, so we will need an adjustment for this amount. In Year +2, the first-iteration projections indicate that assets will exceed liabilities and equities by a total of $1,137.2 million, so we need an incremental adjustment of $16.5 million in Year +2, and so on. We could use a number of flexible financial accounts for this adjustment each year, depending on **Starbucks'** strategy for investments and capital structure. Consider the following options:

- Increase short-term and/or long-term investment securities, assuming **Starbucks** will retain excess capital in marketable securities for financial flexibility.
- Reduce short-term or long-term debt, assuming **Starbucks** will use its financial flexibility to reduce leverage.
- Reduce retained earnings by increasing projected dividends.
- Reduce retained earnings by increasing common share repurchases.

Given the fact that **Starbucks** has clearly demonstrated its willingness and ability to pay out increasing amounts of capital to shareholders through dividends and share repurchases, we adjust share repurchases as the flexible financial account. Equivalently, we could assume that **Starbucks** will distribute the excess capital to shareholders through additional dividends rather than treasury stock repurchases. Either assumption, that **Starbucks** will return the excess capital to shareholders through increased dividends and/or common share repurchases, will have the same effect on total assets, total liabilities, total shareholders' equity, and net income. Therefore, in Year +1, we will increase the share repurchase forecast by $1,153.7 million, the amount necessary to balance the balance sheet. This simply means that if **Starbucks'** financial performance and position during Year +1 exactly match our forecasts, then **Starbucks** can use a total of $2,653.7 million (the original forecast of $1,500.0 plus the adjustment of $1,153.7 million) for share repurchases, keeping total assets in balance with total liabilities and equity. In Years +2 through +5, we will also adjust our share repurchase forecasts of $1,500.0 million up or down each year by the incremental amount of the necessary adjustment to balance the balance sheet (that is, –$16.5 million in Year +2, $98.2 million in Year +3, and so on). The projected total amounts of share repurchases are as follows:

Common Share Repurchases	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Original projections of share repurchases	$1,500.0	$1,500.0	$1,500.0	$1,500.0	$1,500.0
Financial flexible adjustments	1,153.7	(16.5)	98.2	257.6	9.9
Adjusted total share repurchases	$2,653.7	$1,483.5	$1,598.2	$1,757.6	$1,509.9

After adjusting our projected share repurchases as necessary to balance the balance sheet, the Retained Earnings projections are as follows (allow for rounding):

Retained Earnings	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Beginning balance	$ 5,974.8	$ 4,927.8	$ 5,215.6	$ 5,624.1	$ 6,131.9
Projected net earnings	2,794.3	3,080.6	3,490.0	3,939.8	4,895.0
Projected dividend payments	(1,187.6)	(1,309.2)	(1,483.2)	(1,674.4)	(2,080.4)
Projected common share repurchases	(2,653.7)	(1,483.5)	(1,598.2)	(1,757.6)	(1,509.9)
Ending balance	$ 4,927.8	$ 5,215.6	$ 5,624.1	$ 6,131.9	$ 7,436.6

The final projections of the balance sheet total amounts, which you should verify by referring back to the projected balance sheets presented in Exhibit 10.4, are as follows (allow for rounding):

Projected Balance Sheet Amounts	Year +1	Year +2	Year +3	Year +4	Year +5
Total assets (A)	$12,829.0	$13,540.1	$14,495.5	$15,597.5	$17,552.1
Total liabilities	$ 8,056.7	$ 8,477.5	$ 9,021.1	$ 9,611.6	$10,254.8
Common stock and additional paid-in capital	43.9	46.3	49.6	53.4	60.1
Accumulated other comprehensive loss	(199.4)	(199.4)	(199.4)	(199.4)	(199.4)
Retained earnings	4,927.8	5,215.6	5,624.1	6,131.9	7,436.6
Total liabilities and equity (L + E)	$12,829.0	$13,540.1	$14,495.5	$15,597.5	$17,552.1
Difference: A – (L + E)	$ 0.0	$ 0.0	$ 0.0	$ 0.0	$ 0.0

Closing the Loop: Solving for Codetermined Variables

If we had added the excess capital to interest-earning asset accounts (for example, investment securities or cash) or subtracted it from interest-bearing short-term or long-term debt, the projected amounts for interest income or interest expense would have to be adjusted on the income statement. This would have created an additional set of codetermined variables in the financial statement forecasts.

For example, suppose we use long-term debt as the flexible financial account and adjust it to balance assets with liabilities and shareholders' equity. To determine the necessary adjustment to long-term debt, we must first forecast all of the other asset, liability, and shareholders' equity amounts, including retained earnings. To forecast retained earnings, we must forecast net income, which depends on interest expense on long-term debt. To determine retained earnings, we must also forecast dividends, which depend on net income. Thus, it is necessary to solve simultaneously for at least five variables.

This problem might seem intractable, but it is not because of the computational capabilities of computer spreadsheet programs such as Excel. To solve for multiple variables simultaneously, check the box in Excel to "Enable iterative calculation" and allow for up to 1,000 iterations. (This box appears in different places in different versions of Excel. In recent versions, it appears within the Formulas menu under the Options tab under the File tab.) Excel will then solve and resolve circular references up to 1,000 times until all calculations fall within the specified tolerance for precision. Then you can program each cell to calculate the variables needed, even if they are simultaneously determined with other variables. The default settings in FSAP allow for iterative simultaneous computations, but some versions of Excel automatically reset the default settings. Follow these steps to double-check that the FSAP spreadsheet will compute codetermined variables simultaneously.

FSAP

Step 7: Project the Statement of Cash Flows

LO 10-2g

Build forecasts of future balance sheets, income statements, and statements of cash flows by applying the seven-step forecasting framework to project: g. cash flows.

The final step of the seven-step forecasting process involves projecting the statement of cash flows, which we described in Chapter 3. This is a relatively straightforward task because the statement of cash flows simply characterizes all of the changes in the balance sheet in terms of the implications for cash. Thus, we derive the implied cash flows directly from the projected changes in all of the balance sheet accounts (other than the cash account, of course), as follows:

- Increases in assets imply uses of cash.
- Decreases in assets imply sources of cash.
- Increases in liabilities and shareholders' equity imply sources of cash.
- Decreases in liabilities and shareholders' equity imply uses of cash.

Tips for Forecasting Statements of Cash Flows

Here are a few coaching tips for preparing forecasts of statements of cash flows:

- The statement of cash flows will only reconcile with the projected income statement and balance sheets when the balance sheets balance and the income statement articulates with the balance sheets. (That is, net income is included in retained earnings.)
- You should *not* attempt to project future statements of cash flows from historical statements of cash flows. Unfortunately, unlike historical balance sheets and income statements, historical statements of cash flows *do not* provide good bases for projecting future cash flows because many of the line items on the statement of cash flows are difficult to reconcile with historical changes in balance sheet amounts. The reason is that in preparing the statement of cash flows, the accountant can aggregate or disaggregate numerous cash flows on each line item of the statement and you may not be able to determine what amounts have been aggregated or disaggregated. For example, the accountant must report separately the net cash flow implications of a business acquisition on one line of the statement, but the business acquisition causes changes in many asset and liability accounts. In addition, the accountant may choose to disclose details of cash flows that you cannot verify. For example, the accountant might disclose separately in the statement of cash flows the amounts of marketable securities purchased and sold, but you cannot verify those amounts because you can only observe the net change in the marketable securities from the beginning to the end of the year.
- We strongly recommend simply following the steps below to compute the *implied statement of cash flows* from the projected balance sheets and income statements, which you can observe and verify. We have programmed the Forecasts spreadsheet in FSAP (Appendix C) to automatically calculate implied statements of cash flows from the projected income statements and balance sheets.

FSAP

Specific Steps for Forecasting Implied Statements of Cash Flows

Exhibit 10.5 presents the projected implied statement of cash flows for **Starbucks** for 2015 and for Years +1 through +6. Line item numbers are in the right-most column of the exhibit, and we describe the derivation of the amounts on each line item next. You should verify how the projected implied statements of cash flows in Exhibit 10.5 capture the cash inflows and outflows described in each of the following line items.

(1) **Net Income:** Use the amounts in the forecasted income statements (Exhibit 10.3).

(2) **Depreciation Expense:** Add back the projected amount of depreciation expense included in net income. The depreciation expense forecast should reconcile with the change in accumulated depreciation on property, plant, and equipment on the projected balance sheet (minus the decrease in accumulated depreciation from assets that were sold or retired, if any).

Exhibit 10.5

Starbucks
Actual and Forecasted Statements of Cash Flows
(Actual and forecasted amounts in bold. Amounts in millions; allow for rounding.)

	Actuals	Forecasts						
IMPLIED STATEMENTS OF CASH FLOWS	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6	line
Net Income	$ 2,759.3	$ 2,794.3	$ 3,080.6	$ 3,490.0	$ 3,939.8	$ 4,895.0	$ 5,041.8	(1)
Add back depreciation expense (net)	491.4	1,104.2	1,254.2	1,434.2	1,644.2	1,151.6	364.3	(2)
Add back amortization expense (net)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(3)
<Increase> Decrease in receivables—net	(88.0)	(106.0)	(82.8)	(107.7)	(117.9)	(129.1)	(37.9)	(4)
<Increase> Decrease in inventories	(215.5)	(21.3)	(113.1)	(145.2)	(156.8)	(169.3)	(57.4)	(5)
<Increase> Decrease in prepaid expenses	(48.6)	(17.8)	(17.8)	(17.8)	(17.8)	(17.8)	(12.7)	(6)
Increase <Decrease> in accounts payable—trade	150.5	(18.7)	63.6	75.1	79.3	85.6	29.1	(7)
Increase <Decrease> in current accrued liabilities	246.3	231.1	183.0	241.1	263.3	287.3	89.0	(8)
Increase <Decrease> in insurance reserves	28.7	6.7	6.9	7.2	7.4	7.6	7.8	(9)
Increase <Decrease> in stored value card liabilities	189.3	129.2	102.3	134.7	147.1	160.6	49.7	(10)
Net change in deferred tax assets and liabilities	10.0	464.7	74.6	67.1	60.4	54.4	(14.7)	(11)
Increase <Decrease> in long-term accrued liabilities	233.1	82.1	65.0	85.6	93.5	102.0	31.6	(12)
Net Cash Flows from Operations	$ 3,756.5	$ 4,648.5	$ 4,616.6	$ 5,264.2	$ 5,942.4	$ 6,428.0	$ 5,490.7	(13)
<Increase> Decrease in property, plant, & equip. at cost	(1,060.7)	(1,400.0)	(1,500.0)	(1,800.0)	(2,100.0)	(2,400.0)	(565.3)	(14)
<Increase> Decrease in short-term investments	54.1	(2.4)	(2.5)	(2.6)	(2.7)	(2.7)	(2.8)	(15)
<Increase> Decrease in long-term investments	168.8	(27.0)	(28.1)	(29.3)	(30.6)	(31.9)	(24.3)	(16)
<Increase> Decrease in amortizable intangible assets (net)	(246.9)	(26.0)	(27.3)	(28.7)	(30.1)	(31.6)	(19.9)	(17)
<Increase> Decrease in goodwill and nonamort. intangibles	(719.2)	(78.8)	(82.7)	(86.8)	(91.2)	(95.7)	(60.3)	(18)
<Increase> Decrease in other assets	(217.0)	(20.8)	(21.8)	(22.9)	(24.1)	(25.3)	(15.9)	(19)
Net Cash Flows from Investing Activities	$(2,020.9)	$(1,555.0)	$(1,662.5)	$(1,970.4)	$(2,278.7)	$(2,587.3)	$ (688.6)	(20)

Exhibit 10.5 (Continued)

	Actuals	Forecasts						
IMPLIED STATEMENTS OF CASH FLOWS	2015	Year +1	Year +2	Year +3	Year +4	Year +5	Year +6	line
Increase <Decrease> in short-term debt	0.0	400.0	(400.0)	350.0	(350.0)	0.0	0.0	(21)
Increase <Decrease> in long-term debt	299.2	600.0	400.0	(350.0)	350.0	0.0	100.4	(22)
Increase <Decrease> in common stock + paid-in capital	2.5	1.3	2.4	3.3	3.8	6.7	1.8	(23)
Increase <Decrease> in accum. OCI	(224.7)	0.0	0.0	0.0	0.0	0.0	0.0	(24)
Dividends and share repurchases	(1,989.2)	(3,841.3)	(2,792.8)	(3,081.5)	(3,432.0)	(3,590.3)	(4,824.7)	(25)
Increase <Decrease> in noncontrolling interests	(1.8)	(1.8)	0.0	0.0	0.0	0.0	0.0	(26)
Net Cash Flows from Financing Activities	$(1,914.0)	$(2,841.7)	$(2,790.4)	$(3,078.2)	$(3,428.2)	$(3,583.6)	$(4,722.5)	(27)
Net Change in Cash	$ (178.4)	$ 251.8	$ 163.7	$ 215.6	$ 235.5	$ 257.0	$ 79.6	(28)
Beginning Cash Balance	$ 1,708.4	$ 1,530.0	$ 1,781.8	$ 1,945.5	$ 2,161.2	$ 2,396.7	$ 2,653.7	(29)
Ending Cash Balance	$ 1,530.0	$ 1,781.8	$ 1,945.5	$ 2,161.2	$ 2,396.7	$ 2,653.7	$ 2,733.3	(30)
Check figure								
Net change in cash – Change in cash balance	0	0	0	0	0	0	0	

Note: We label the statement of cash flows and amounts for 2015 as "Actual" even though we derive it from the actual balance sheet and income statement amounts in Starbucks' financial statements rather than from the financial statement forecasts. Appendix A presents the actual statements of cash flows for 2013, 2014, and 2015 as reported by Starbucks according to U.S. GAAP.

(3) Amortization Expense: Add back amortization expense on amortizable intangible assets.[21] The amount of amortization expense to add back to net income should reconcile with the change in amortizable intangible assets balance, adjusted for any new investments in those assets (which should be included as cash outflows in the investing section of this statement). For some firms, if the amount of amortization expense is not large, you can ignore adding it back to net income to compute cash flow from operating activities and simply include the net change in amortizable intangible assets in the investing section. This will slightly understate cash inflows from operations and slightly understate cash outflows for investing activities, but the two effects will offset so that net cash flow is not affected.

(4)–(10) Working Capital Accounts: Adjust net income for changes in all of the various operating current asset and current liability accounts other than cash (such as accounts receivable, inventory, accounts payable, accrued expenses, and others) appearing on the projected balance sheets.

(11), (12) Deferred Taxes and Long-Term Accrued Expenses: Adjust net income for changes in deferred taxes, noncurrent liabilities for accrued expenses, and changes in other noncurrent liabilities. These items include changes in long-term accruals for expenses that are part of operations, including deferred taxes, pension and retiree benefit obligations, warranties, and other noncurrent liabilities that appear on the projected balance sheets.

(13) Net Cash Flows from Operations: The sum of lines (1) through (12) is the implied amount of net cash flows from operating activities.

(14) Property, Plant, and Equipment: The amount on this line captures cash outflows for the projected capital expenditures included in the change in property, plant, and equipment (at cost) on the projected balance sheet in Exhibit 10.4, minus any cash proceeds from sales of property, plant, and equipment. As a check, make sure the statement of cash flows captures all of the net cash flow implications of property, plant, and equipment. To verify this, the amount of depreciation expense added back to net income *minus* cash outflows for capital expenditures *plus* cash inflows for any asset sales or retirements should equal the change in net property, plant, and equipment on the projected balance sheet.

(15), (16) Investment Securities (net): The statement of cash flows classifies net purchases and sales of short-term and long-term investment securities as investing transactions. The net changes in these accounts on the projected balance sheets determine the cash flow amounts for these items on the statement of cash flows. Some error in the implied cash flow amount from investment securities can occur. This change should be increased (become less negative) for the excess (if any) of equity earnings over dividends received from noncontrolled affiliates (which is a noncash increase in this asset amount). Similarly, the excess of equity earnings over dividends received also should be subtracted from net income in the operating section of the statement of cash flows. Rather than making assumptions about this relatively immaterial item (the effects of which completely offset each other), we simply treat the entire change in investments as an investing transaction. This choice means that cash flows from operating activities may be slightly overstated and cash flows from investing activities may be slightly understated by an equivalent amount but that the net change in cash each year is not affected.

(17) **Amortizable Intangible Assets:** Enter the implied net cash flows from any changes in amortizable intangible assets on this line, which should include any projected cash outflows to acquire amortizable intangible assets, net of any cash inflows from sales or retirements of such assets. As discussed in Item (3), we add back amortization expense to net income in the operating section of the statement of cash flows. Thus, our adjustment for cash outflows or inflows for amortizable intangible assets in the investing section of the statement should not include the effects of amortization expense. Given that amortizable intangibles are commonly shown on balance sheets net of accumulated amortization, the change in the net amortizable intangible assets account balance will reflect both effects: cash flows from investing activities and amortization expense. To isolate the cash flows from investing, you should add amortization expense back to the net change in this account balance.

(18) **Goodwill and Indefinite-Lived (Nonamortizable) Intangible Assets:** Enter the implied cash flows from changes in goodwill and indefinite-lived intangible assets on this line. These assets are not amortized, so the net change on the projected balance sheets should reflect cash outflows to acquire new goodwill and nonamortizable intangible assets, net of any cash inflows from selling or retiring such assets. If the account balance for goodwill or nonamortizable intangible assets has declined because of an impairment charge, you should add this noncash charge back to net income in the operating section of the statement of cash flows and adjust accordingly the cash flow implications from net changes in goodwill or nonamortizable intangibles in the investing section (similar to adding back amortization expense).

(19) **Other Noncurrent Assets:** Enter the changes in other noncurrent assets on this line. The changes in the other noncurrent asset accounts on the projected balance sheets measure the cash outflows to acquire such assets net of any cash inflows from sales or retirements of such assets.

(20) **Net Cash Flows from Investing Activities:** The sum of lines (14) through (19) on Starbucks' projected implied statement of cash flows measures the implied amount of net cash flows from investing activities.

(21), (22) **Short-Term and Long-Term Debt:** Changes in interest-bearing debt (short-term notes payable, current maturities of long-term debt, and long-term debt) on the projected balance sheets are financing activities.

(23) **Changes in Common Stock and Additional Paid-in Capital:** These amounts represent the financing cash flows from changes in the common stock and paid-in capital accounts on the projected balance sheets.

(24) **Changes in Accumulated Other Comprehensive Income:** These amounts represent the changes in the accumulated other comprehensive income account that is a component of shareholders' equity on the projected balance sheets.

(25) **Dividends and Common Share Repurchases:** The amounts represent the net cash flow implications of dividend payments as well as common share repurchases that are captured in the net change in retained earnings (or in the treasury stock account) on the projected balance sheets.

(26) **Noncontrolling Interests:** Enter the projected amounts of dividends to noncontrolling interests as cash outflows, net of any cash inflows (if any) from additional investments by noncontrolling investors.

(27) Net Cash Flows from Financing Activities: The sum of lines (21) through (26) measures the implied amount of net cash flows from financing activities.

(28) Net Change in Cash: The aggregate of the amounts of cash flows from operating activities (line 13), investing activities (line 20), and financing activities (line 27). This total should equal the change in cash on the projected balance sheets.

(29), (30) Beginning and Ending Balance in Cash: The net change in cash on the statement of cash flows (line 28) should equal the change in cash on the projected balance sheets.

Shortcut Approaches to Forecasting

LO 10-3

Understand how and when to use shortcut forecasting techniques.

This chapter emphasizes and demonstrates a methodical, detailed approach to forecasting drivers of expected future operating, investing, and financing activities that will determine future amounts for individual accounts on the income statement, balance sheet, and statement of cash flows. In some circumstances, however, you may find it necessary to forecast income statement and balance sheet totals directly without carefully considering each account. This shortcut approach has the potential to introduce forecasting error if the shortcut assumptions do not fit each account very well. On the other hand, if the firm is stable and mature in an industry in steady-state equilibrium, shortcut forecasting techniques are efficient approaches to project current steady-state conditions to the future. This section briefly illustrates shortcut approaches for forecasting **Starbucks'** income statements and balance sheets.

Projected Revenues and Income Approach

Shortcut projections for total revenues and net income can be developed using **Starbucks'** recent revenue growth rates and net profit margins. The common-size and rate-of-change income statement data discussed in Chapter 1 reveal that, from 2011 through 2015, **Starbucks** generated a compound annual growth rate in revenues of 12.3% and an average net profit margin of 9.6%. If we simply use these ratios to forecast revenues and net income over Years +1 to +5, the projected amounts are as follows:

Shortcut Projections	2015	Year +1	Year +2	Year +3	Year +4	Year +5
Total net revenues	\$19,162.7	\$21,519.7	\$24,166.6	\$27,139.1	\$30,477.2	\$34,225.9
Compounded annual growth	12.3%	12.3%	12.3%	12.3%	12.3%	12.3%
Net earnings	\$ 2,759.3	\$ 2,065.9	\$ 2,320.0	\$ 2,605.4	\$ 2,925.8	\$ 3,285.7
Historical average profit margin	9.6%	9.6%	9.6%	9.6%	9.6%	9.6%

These shortcut projections for revenues are much higher and the projections for net income are much lower than the detailed revenue and income projections developed for **Starbucks** throughout this chapter. The more detailed projections capture specific factors driving growth in revenues, as well expected changes in individual expenses relative to revenues. The shortcut approach simply assumes that existing relations between revenues and expenses will persist linearly into the future.

Projected Total Assets Approach

For the balance sheet, total assets can be projected using the recent historical growth rate in total assets. Between fiscal 2011 and 2015, **Starbucks'** total assets grew at a compound annual rate of 14.3%. Over that same period, total liabilities averaged 48.7% of total assets, and shareholders' equity averaged 51.3%. If a shortcut approach assumes this historical growth rate and capital structure will continue through Year +5, then projected total assets, liabilities, and equities will be as follows (allow for rounding):

Shortcut Projections	**2015**	**Year +1**	**Year +2**	**Year +3**	**Year +4**	**Year +5**
Total assets	$12,446.1	$14,225.9	$16,260.2	$18,585.4	$21,243.1	$24,280.9
Compounded annual growth	14.3%	14.3%	14.3%	14.3%	14.3%	14.3%
Total liabilities	$ 6,626.3	$ 6,928.0	$ 7,918.7	$ 9,051.1	$10,345.4	$11,824.8
Average as a percent of assets	48.7%	48.7%	48.7%	48.7%	48.7%	48.7%
Total shareholders' equity	$ 5,819.8	$ 7,297.9	$ 8,341.5	$ 9,534.3	$10,897.7	$12,456.1
Average as a percent of assets	51.3%	51.3%	51.3%	51.3%	51.3%	51.3%

Using common-size balance sheet percentages to project individual assets, liabilities, and shareholders' equity encounters (at least) three potential shortcomings. First, in using common-size percentages for forecasting, you assume the firm maintains a constant mix of assets, liabilities, and equities, and that each asset, liability, and equity account grows at the same rate as total assets. This strong assumption may not be valid for a majority of firms, which tend to experience dynamic, nonlinear changes over time. Second, using the common-size percentages does not permit you to change the assumptions about the future behavior of an individual asset or liability. For **Starbucks**, for example, we forecast continued, substantial increases in dividend payments and share repurchases. Third, the common-sized shortcut approaches do not allow you an effective mechanism to forecast major transactions that reflect disproportionate shifts in assets relative to liabilities and equities.

In general, shortcut approaches to forecasting have the benefit of greater efficiency but greater potential for error as compared to more thoughtful and deliberate forecasts of income statement and balance sheet accounts. Given that forecast errors can be very costly if they lead to bad investment decisions, we strongly advocate the careful, detailed approach to projecting financial statements by forecasting the firm's future operating, investing, and financing activities using individual income statement and balance sheet accounts.